The Economic Effects of Decentralization: Firm Performance and Fiscal Capacity

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October 15, 2026

Abstract

Decentralization is widely argued to improve public service delivery, yet its effects on private-sector activity and fiscal capacity remain poorly understood. We study Uganda's 2015/16 creation of 23 new districts using administrative tax records in a staggered difference-in-differences design. District creation increased firm revenues by an average of 13 percent over the study period, with effects emerging two years after the reform and growing thereafter. Firms increase expenditures on intermediate inputs and labor, while profits do not significantly change. Stronger local demand and improved infrastructure, including more reliable electricity, help explain these gains. Effects are concentrated in relatively symmetric district splits, with no detectable gains in highly asymmetric splits. Decentralization also strengthened central fiscal capacity by increasing corporate income tax remittances, effective tax rates, and labor income tax revenue. Overall, decentralization can stimulate private-sector activity, and strengthen central revenue mobilization, but institutional design matters.

JEL codes: H77, H71, H25, O17, D22
Key Words: Decentralization, District Creation, Firm Performance, State Capacity.

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I. INTRODUCTION

Developing countries have increasingly embraced decentralization of governance, often through the subdivision of existing administrative units into smaller jurisdictions. The principal justification for this trend is that bringing government closer to citizens can improve the efficiency and responsiveness of public goods and service provision (Bardhan 2002).[1] Recent empirical work provides support for this view, documenting improvements in public service delivery following decentralization across a range of settings (Singhania 2022; Dahis and Szerman 2024; Narasimhan and Weaver 2024).[2] Yet two important economic consequences of these reforms remain less understood. First, it is unclear whether improvements in local public goods and services translate into tangible gains for the private sector. Anecdotal accounts have long emphasized decentralization as an important institutional contributor to China's industrial development (Bardhan 2002), but rigorous firm-level evidence, particularly from low-income countries, remains scarce. Second, decentralization may alter the fiscal capacity of the state itself, affecting its ability to mobilize revenue even when major taxes remain under central government authority. Understanding both margins is important for evaluating decentralization, particularly as concerns about the administrative costs of creating additional layers of government have grown in countries already facing severe fiscal constraints.

To the extent that decentralization improves the provision of productivity-enhancing public goods, such as electricity, transport infrastructure, and regulatory services, it may improve the local business environment and raise firm performance. Several countervailing forces, however, make this effect theoretically ambiguous. Smaller administrative units may sacrifice economies of scale in public goods provision (Alesina and Spolaore 1997; Grossman et al. 2017; Cassidy and Velayudhan 2026). Reduced political competition in smaller jurisdictions may weaken leadership quality and incentives to perform (Gerring et al. 2015), while greater information frictions may undermine voters' ability to hold local officials accountable (Boffa et al. 2016). Newly created governments may also lack the bureaucratic capacity required to perform their expanded responsibilities effectively (Grossman et al. 2017; Cassidy and Velayudhan 2026). Whether greater administrative proximity ultimately improves the environment in which firms operate therefore depends on whether gains from responsiveness and proximity outweigh losses from fragmentation, weaker political incentives, and limited local capacity.

[1] Decentralization is also viewed as a means of diffusing social and political tensions and preserving local cultural and political autonomy (Bardhan 2002).

[2] Decentralization may also increase the responsiveness of public investment to local needs (Faguet 2004), foster intergovernmental competition, and strengthen checks and balances (Bardhan 2002).

Decentralization may also have important consequences for the fiscal capacity of the central state. Bringing government closer to taxpayers may strengthen the fiscal contract by increasing the visibility and responsiveness of the state, enhancing its legitimacy, and improving taxpayers' willingness to comply (Besley 2020; Weigel 2020). Greater local state presence may also reduce the informational costs of taxation by improving the government's knowledge of local economic activity and facilitating interactions between taxpayers and public authorities (Balán et al. 2022). These mechanisms suggest that the fiscal consequences of decentralization may extend beyond the budgets of newly created local governments. Even when major tax instruments remain under central government authority, decentralization may affect central revenue collection by shaping both taxpayers' compliance and the size of the formal tax base. Understanding these fiscal spillovers is therefore important for assessing the broader consequences of decentralization.

In this paper, we study how the creation of new administrative districts in Uganda affects the performance of local firms and the fiscal capacity of the central government. We focus on the 2015/16 wave of district creation, in which 23 new districts were carved out of existing ones. Uganda has experienced several episodes of administrative fragmentation, including a large wave in 2009/10 that created 33 new districts. We study the 2015/16 episode because our firm-level administrative data begin in 2013/14, providing multiple pre-reform years with which to implement a difference-in-differences design and assess pre-treatment dynamics.

Uganda provides a particularly well-suited setting for studying these questions for three reasons. First, districts exercise substantial administrative, fiscal, and political authority under Uganda's Local Governments Act, so district creation represents more than a redrawing of administrative boundaries: it creates new centers of local government with meaningful responsibility for public service delivery and local administration (Section II provides institutional details). Second, administrative tax records covering the universe of firms registered for corporate income tax, combined with household survey data on local public goods and services, allow us to trace both the effects of district creation on firm outcomes and potential mechanisms through which these effects operate. Third, while district governments play an important role in local administration and service delivery, major taxes on firms and workers, including corporate income tax and employee income tax (PAYE), remain under central government authority. This institutional structure allows us to examine whether decentralization generates fiscal spillovers across levels of government by affecting the central government's ability to mobilize revenue from firms and formal employment.

To identify the causal effects of district creation, we exploit variation in the timing

of district splits in a staggered difference-in-differences design. The approach compares changes in outcomes for firms exposed to district splitting with contemporaneous changes among firms not yet exposed or unaffected by the reform, while allowing the effects of decentralization to evolve over time. Multiple years of pre-reform administrative data allow us to examine pre-treatment dynamics and assess the plausibility of the parallel-trends assumption.

We first show that district splitting substantially increases the fiscal resources allocated to affected local governments. Total intergovernmental grant allocations per capita rise by approximately 11 percent. Because we aggregate allocations to the parent and newly created child districts after a split, these estimates capture additional resources flowing to the affected territory rather than a mechanical division of existing allocations across administrative units. The increase is particularly pronounced for development grants, which rise by approximately 23 percent, compared with about 8 percent for recurrent grants. This pattern suggests that district splitting expands the resources available for local investments that may affect the local business environment, rather than merely generating additional administrative overhead. Crucially, the increase in grant allocations to split districts does not reflect a reallocation from unsplit districts, but rather an expansion in the intergovernmental transfer envelope.

Turning to local firm activity, the reform increased firm revenues by approximately 13 percent on average, with effects emerging two years after district splitting and growing steadily thereafter. Despite the significant increase in revenues, the reform had no significant effect on profits, as the revenue gains are accompanied by proportional increases in expenditures on intermediate inputs and labor, as well as an expansion in formal employment and the wage bill. This pattern suggests that decentralization expands the scale of local activity, with firms purchasing more inputs and producing more output, rather than improving their profitability on the margin.[3] We provide suggestive evidence that increased local demand and improved infrastructure help explain these effects: revenue gains are statistically significant among firms in non-tradable sectors, while electricity availability increases by approximately 13 percent following district splitting, consistent with greater local capacity to provide infrastructure essential for firm production.

We provide new evidence showing that the gains from decentralization depend critically on how the boundaries are drawn. Revenue gains are concentrated among firms in districts that were split symmetrically, while highly asymmetric splits generate no detectable effects. This finding suggests that the benefits of decentralization depend crit-

[3]Because we observe only four post-reform periods, we are unable to assess whether effects on firm profitability emerge over a longer horizon.

ically on whether the resulting jurisdictions retain sufficient scale to support effective administration and public goods provision. Highly asymmetric splits, which may arise when district creation is motivated by reasons other than administrative considerations, do not generate the same gains for local firms.

Finally, we show that decentralization generates fiscal gains that extend beyond the newly created local governments. Corporate income tax (CIT) remittances increase by approximately 23 percent relative to its pre-reform mean, while the effective tax rate rises by about 2 percentage points. Because profits respond only weakly, the joint increase in CIT payments and effective tax rates is difficult to explain through an expansion of the corporate tax base alone and is consistent with an improvement in effective tax collection. We provide evidence that firms defer less of their tax liabilities following the reform. We provide evidence consistent with firms recognizing a larger share of their tax obligations in the current period rather than deferring them to future periods. Decentralization also increases labor income tax revenue collected through the Pay-As-You-Earn (PAYE) withholding system. This latter response coincides with increases in formal employment and the formal wage bill, pointing to an expansion of the formal labor tax base. Combining CIT and PAYE, centrally collected income-tax revenues increase by approximately 7.5 percent. Thus, decentralization appears to strengthen central fiscal capacity through two complementary margins: greater effective taxation of corporate income and an expansion of taxable formal employment.

These findings contribute to our understanding of decentralization along two dimensions. First, they provide firm-level evidence that improvements in local governance can translate into economically meaningful changes in private-sector activity, extending a literature that has largely focused on public service delivery and household outcomes. Second, they show that the fiscal consequences of decentralization need not be confined to subnational governments: greater local state presence can generate positive fiscal spillovers for the central government by expanding formal economic activity and strengthening effective revenue collection. At the same time, the sharp heterogeneity by the configuration of district splits underscores that these gains are not an automatic consequence of creating smaller jurisdictions. More broadly, the results suggest that the economic returns to decentralization depend on balancing the benefits of administrative proximity against the need to preserve sufficient scale for effective local governance.

This paper contributes to two related strands of literature. First, it contributes to the literature on decentralization and the provision of public goods. While a few studies find that subdividing administrative jurisdictions into smaller units can harm public goods provision (Lewis 2017; Cohen 2024), the weight of recent evidence points in the op-

posite direction, documenting improvements in public goods delivery (Singhania 2022; Narasimhan and Weaver 2024; Dahis and Szerman 2024), greater responsiveness of public investment to local needs (Faguet 2004), and gains across a broad range of development outcomes (Dahis and Szerman 2024). This literature, however, falls short of asking whether these improvements in public goods translate into tangible economic gains for the private sector. We extend this line of inquiry by examining whether decentralization improves the provision of public goods in a manner that supports the performance of local firms, a link that is central to the economic case for decentralization but has not been empirically established.

Second, the paper contributes to the literature on the relationship between government structure and economic growth. Several studies find that decentralization enhances subnational economic growth (Stansel 2005; Hatfield 2015; Zhang et al. 2019), while Cassidy and Velayudhan (2026) show that government fragmentation reduces district GDP in the short run in Indonesia. This literature, however, relies on aggregate measures of economic activity, which can document whether growth occurs but not how it arises at the firm level. We contribute by leveraging firm-level panel data covering the universe of firms registered for corporate income tax in Uganda, which allows us to trace the sources of local economic growth, through revenues, costs, profits, and input use, rather than simply measuring its aggregate magnitude. The paper also provides evidence from a low-income country context, where the institutional conditions under which decentralization operates differ markedly from the middle- and high-income settings that dominate the existing literature.

The rest of the paper is organized as follows. Section II describes the institutional context. Section III describes the data sources used in the study, and presents the descriptive analysis. Section IV describes the identification strategy. Section V presents and discusses the empirical results. Section VI presents the conclusion.

II. CONTEXT

II.A. Districts as Local Governments in Uganda

Uganda operates a decentralized system of government in which the district is the principal unit of local government. The 1995 Constitution establishes the district as the basis of the local government system, with lower local governments and administrative units organized beneath it. In rural areas, these include subcounties and, below them, parishes and villages. The decentralization framework assigns local governments responsibility for planning and implementing policies within their jurisdictions and is intended

to promote local participation and accountability in public decision-making (Republic of Uganda 1995; 1997).

Each district has both a political and an administrative structure. The district council is the highest political authority within the district and exercises legislative and executive powers. It is headed by a district chairperson (LC5), who is directly elected by universal suffrage, and includes elected representatives from electoral areas within the district. The administrative arm is led by the Chief Administrative Officer (CAO) appointed by the central government, who serves as the head of the district public service and the district's accounting officer. The CAO is responsible for implementing council decisions, coordinating district departments and lower-level governments, and overseeing the delivery of public services. Districts also maintain specialized technical departments responsible for areas such as finance, planning, engineering, health, education, agriculture, and community development.

District governments have substantial responsibilities for local public service provision. Under the Local Governments Act, these include education, health services, water provision, construction and maintenance of district roads, agricultural extension, land administration and physical planning, trade and business licensing, and community development. Districts are also responsible for preparing integrated development plans that consolidate plans from lower levels of government. National policy in areas such as education, health, agriculture, taxation, and infrastructure remains the responsibility of the central government, but districts play an important role in translating these policies into local service delivery.

District activities are financed through a combination of intergovernmental transfers and locally generated revenue. The Constitution provides for unconditional grants to finance decentralized services, conditional grants earmarked for agreed programs, and equalization grants targeted toward relatively disadvantaged local governments. Districts may also raise own-source revenue through rates, licenses, fees, rents, and other locally assigned taxes. In practice, however, local governments rely heavily on transfers from the central government, with own-source revenues accounting for a relatively small share of their budgets (World Bank 2024). Importantly, for our analysis, major income taxes, including corporate income tax and labor income taxes, remain under central government authority. Thus, changes in district governance can affect centrally collected revenues without altering the formal administration of these taxes.

II.B. District Creation in Uganda

District creation has been a recurring feature of Uganda's administrative structure. At independence in 1962, the country was administered through a small number of large districts inherited from the colonial administration. By 1995, when the current constitution was promulgated, Uganda had 39 districts. This number rapidly increased onward reaching 135 by 2020 (Figure 1), making Uganda a prominent example of subnational administrative proliferation (Green 2010; Grossman and Lewis 2014).

The wave central to our empirical analysis was approved by Parliament in September 2015, when 23 additional districts were authorized to become operational in phases. The first districts from this wave became effective on July 1, 2016, with subsequent districts commencing in later fiscal years. Figure 2 shows the different phases in which resulting fragments become operational.

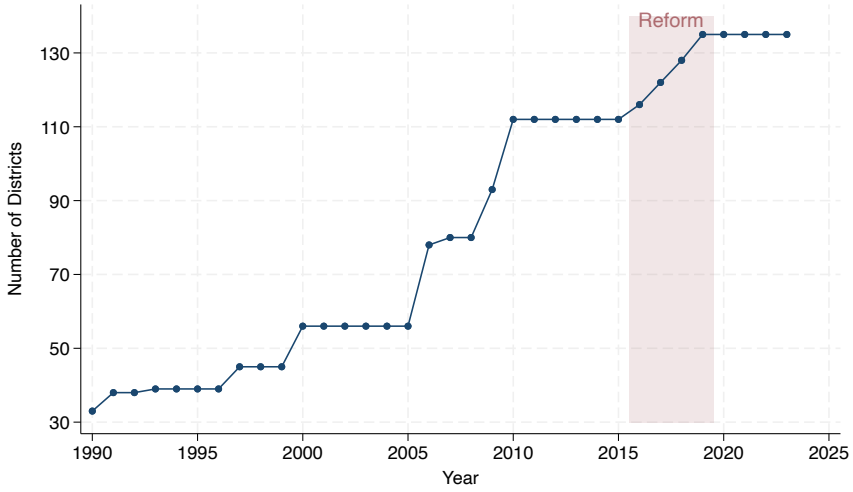


Figure 1: Number of Administrative Districts in Uganda

Notes: The figure plots the number of administrative districts in Uganda from 1990 to 2023. The shaded region indicates the reform period studied in this paper (2015/16–2018/19), during which 23 new districts were carved from existing ones. Source: author's compilation from Uganda Bureau of Statistics records.

New districts are generally created by partitioning the territory of an existing district, producing a new local government alongside the remaining parent district. The 1995 Constitution gives Parliament authority to alter district boundaries and create new districts, with approval requiring a majority of all Members of Parliament. The Constitution specifies that district creation should be motivated by the need for effective administration and to bring public services closer to citizens, while also allowing consideration of population density, geographical features, communication infrastructure, economic viability, and the

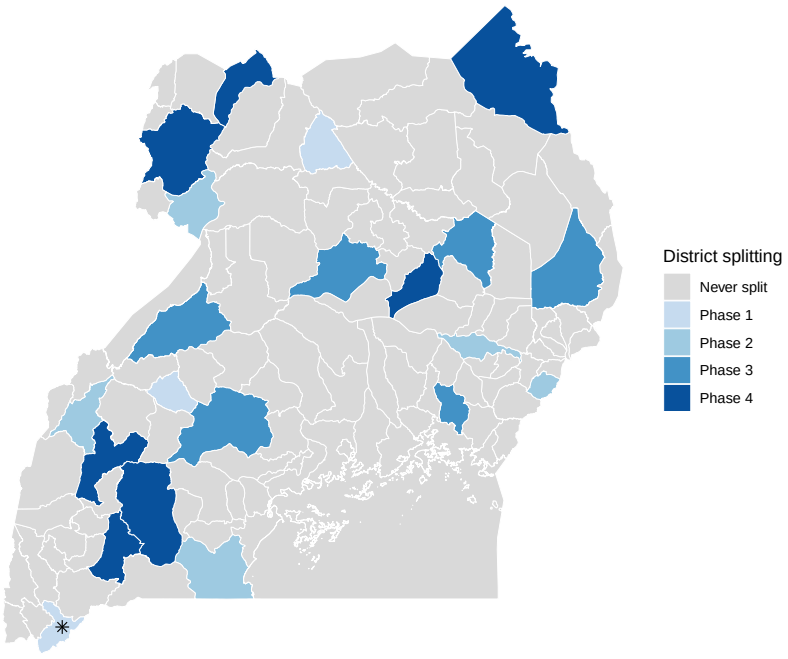


Figure 2: District Splits by Operationalization Phases

Notes: The figure displays Uganda's district boundaries prior to the 2015/16 wave of district creation. Districts affected by the reform are shaded by the phase in which resulting fragments become operational, and unshaded districts were not affected during the study period. Each phase took effect on 1 July at the start of a fiscal year. Phase 1 (FY2016/17) created Kagadi and Kakumiro (from Kibaale), Omoro (from Gulu), and Rubanda (from Kabale). Phase 2 (FY2017/18) created Bunyangabu (from Kabarole), Butebo (from Pallisa), Kyotera (from Rakai), Namisindwa (from Manafwa), Pakwach (from Nebbi), and Rukiga (from Kabale). Phase 3 (FY2018/19) created Bugweri (from Iganga), Kapelebyong (from Amuria), Kasanda (from Mubende), Kikuube (from Hoima), Kwania (from Apac), and Nabilatuk (from Nakapiripirit). Phase 4 (FY2019/20) created Kalaki (from Kaberamaido), Karenga (from Kaabong), Kazo (from Kiruhura), Kitagwenda (from Kamwenge), Madi-Okollo (from Arua), Obongi (from Moyo), and Rwampara (from Mbarara). The asterisk identifies Kabale, which experienced splits in both Phase 1 and Phase 2.

preferences of the affected population (Republic of Uganda 1995).

Although final authority rests with Parliament, proposals for new districts can originate through several channels. In practice, demands may arise through resolutions of district councils, requests from local political representatives, or proposals initiated by the central government. The Ministry of Local Government assesses proposed districts against the constitutional criteria and, following Cabinet consideration, the responsible minister may table a motion before Parliament. A parliamentary committee subsequently examines the proposal and consults affected areas before the proposal is submitted to a final vote (Awortwi 2011; Grossman and Lewis 2014).

Once a split takes effect, the newly created district becomes a separate local government with its own political leadership, administrative structure, planning process, and budget. It therefore requires a new district council and executive, a CAO and district administration, and the technical departments necessary to perform decentralized functions. This institutional separation is important for our setting: district creation does not merely change the geographic label assigned to residents, but creates a new center of local political authority and public administration.

The stated rationale for district creation has consistently emphasized administrative proximity and service delivery. Smaller jurisdictions may reduce the distance between citizens and local officials, improve information about local needs, and allow government services and infrastructure to be more closely targeted to local conditions. These arguments are often accompanied by expectations that new administrative unit will attract investment, generate local employment through the expansion of the district civil service, and stimulate commercial activity around the new district headquarters. At the same time, the political-economy literature emphasizes that the demand for new districts may reflect additional considerations. Grossman and Lewis (2014), for example, document an important role for local political and ethnic marginalization in demands for new administrative units, while Green (2010) argues that national political and electoral incentives have also contributed to district proliferation in Uganda.

III. DATA AND DESCRIPTIVE STATISTICS

III.A. Data Sources and Sample Construction

Intergovernmental transfers.— We compile annual data on intergovernmental grant allocations from the central government to local governments from approved local government budgets published by Uganda's Ministry of Finance, Planning and Economic Development (MoFPED). The data cover fiscal years 2012/13 through 2019/20 and report detailed approved grant allocations for each district. We end the series in 2019/20 because Uganda subsequently transitioned from sector-based to program-based budgeting, changing the classification and reporting structure of local government budgets and limiting comparability with earlier years. The records identify the recipient local government, distinguishing between wage recurrent, non-wage recurrent, government development grants, and externally financed allocations, which allows us to examine which category responds the most when districts get split. We then aggregate these records to construct total annual grant allocations at the district to examine how the total allocation evolves. The data also contains a sectoral disaggregation, allowing us to construct district-level

allocations to specific functions, such as education, health, water and environment, production, administration, among others. We use this disaggregation to examine how the composition of intergovernmental resources changes following district creation.

Firm-level data.— The main analysis draws on firm-level panel data covering the universe of firms registered for corporate income tax in Uganda over a nine-year period (2013/14–2021/22). The data contain detailed information from firms' balance sheets and profit and loss accounts, from which we extract key economic variables including revenues, profits, and costs. Crucially, the data record each firm's location at the sub-county, county, and district level, which allows us to assign treatment status based on whether the firm's district was split during the reform. The data also include information on industry of operation, which we use to control for sector-specific trends. All monetary values are denominated in Uganda shillings (UGX) and converted to constant 2016/17 prices using the consumer price index to ensure comparability over time.

To study the effects on formal employment, we link the firm-level corporate tax data to employee-level (PAYE) records, which cover the universe of formally employed workers. We measure formal employment as the number of workers reported under each firm. Because PAYE records are filed monthly whereas corporate income tax returns are filed annually, we construct annual employment as the average number of formal workers employed by the firm across the twelve months of the year. We also aggregate workers' reported earnings within each firm to construct the annual formal wage bill, which we use to examine the effects of the reform on formal labor compensation.

Sample restrictions.— The main results are estimated on a balanced panel of firms spanning the full 2013/14 – 2021/22 period. The underlying dataset contains between 28,140 and 59,918 firms per year, of which 9,806 are observed in all nine years. We further restrict the sample to firms that reported positive revenues in the pre-reform year, yielding a balanced panel of 69,723 firm-year observations.

Other data.— To examine additional channels through which decentralization affects firms, we use the Uganda National Panel Survey (UNPS), a nationally representative panel implemented by the Uganda Bureau of Statistics with support from the World Bank's Living Standards Measurement Study. We use four waves: 2013/14 and 2015/16, which precede the reform, and 2018/19 and 2019/20, which span the period in which the new districts became operational. Each wave covers roughly 3,100 households and 300 communities. The household questionnaire records information on electricity availability and reliability, including the number of hours of electricity supply per day among connected households, providing one measure of access to productivity-enhancing infras-

tructure. The community questionnaire, completed by local officials, provides additional measures of access to infrastructure and services that are important for firm performance. It records information on the availability of public and private services in each community, including financial institutions, agricultural extension services, vocational schools, and police services, as well as on the stock, condition, maintenance, and construction of roads in the surrounding sub-county. The road measures distinguish between trunk roads maintained by the central government and feeder roads maintained by district governments.

III.B. Descriptive Statistics

Pre-reform district characteristics.— Figure 3 compares pre-split characteristics of districts that were subsequently split (ever-treated) with those that remained unaffected (never-treated) during the study period. Panel A shows that districts that got split were substantially more populous prior to district creation than those that were never split. This pattern is consistent with population pressure being one factor associated with the creation of new districts. Despite this difference in size, Panel B shows that per capita grant allocations from the central government were broadly similar across the two groups and, if anything, somewhat lower among districts that got split. Thus, districts that were subsequently split were larger, but they do not appear to have been systematically more generously funded on a per capita basis before the reform. This distinction is important for interpreting the analysis of intergovernmental grants, which examines whether district creation itself is followed by changes in the resources allocated to local governments.

Firm-level characteristics:—Table 1 reports pre-reform characteristics of firms in the administrative tax data. The firms are heterogeneous in size, as reflected in the large dispersion of assets, revenues, and input expenditures. In the full CIT universe, average annual revenues are UGX 596.6 million and average total assets are UGX 262.4 million, with standard deviations substantially exceeding the corresponding means. The average profit margin is 14 percent, while approximately 20 percent of firms report negative pre-tax profits. Firms employ an average of 5.7 formal workers, although the substantial dispersion in both the formal wage bill and employment indicates considerable heterogeneity in firm scale.

Firms located in districts that are subsequently split are, on average, smaller than firms in never-treated districts prior to the reform. Ever-treated firms report average revenues of UGX 430.9 million, compared with UGX 615.4 million among never-treated firms, and similar differences are evident in assets, cost of sales, employee compensation, and the

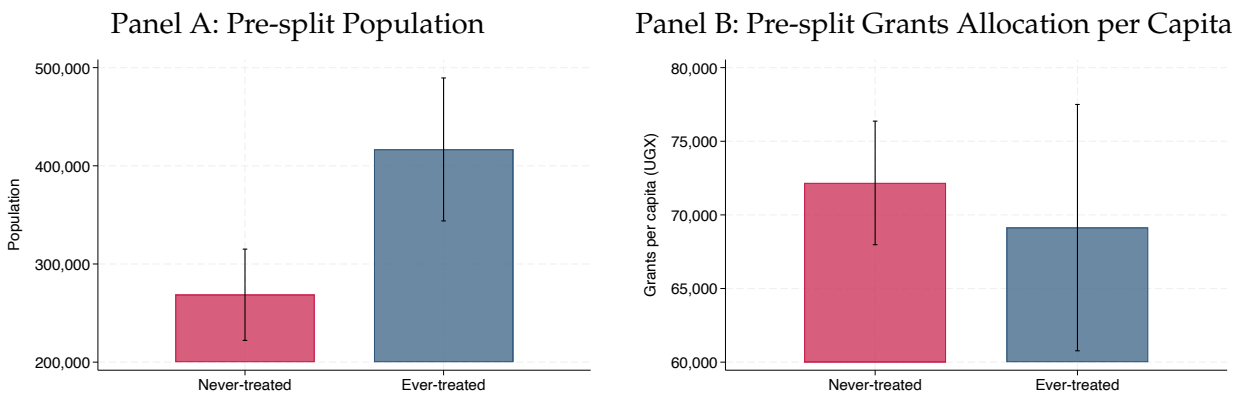


Figure 3: Pre-Reform Population and Grants

Notes: The figure presents group means and 95 percent confidence intervals for pre-reform population and intergovernmental grant allocations, comparing districts that were subsequently split (ever-treated) with districts that remained unaffected during the sample period (never-treated). Panel A reports the average district population from the 2014 Uganda Population and Housing Census. Panel B reports per capita intergovernmental grant allocations from the central government, expressed in Uganda Shillings (UGX) and averaged over the pre-reform years. Grant allocations are drawn from Ministry of Finance, Planning and Economic Development data on district-level central government transfers. Bars represent group means, and whiskers denote 95 percent confidence intervals.

formal wage bill. Consistent with their smaller scale, treated firms also remit less CIT and PAYE prior to district splitting. These differences are considerably smaller for employment: ever-treated firms employ an average of 5.2 formal workers compared with 5.8 among never-treated firms. Treated firms also appear somewhat more profitable prior to the reform, with an average profit margin of 18 percent and 13 percent reporting losses, compared with 14 percent and 20 percent, respectively, among never-treated firms.

Part of the differences in firm scale reflects the composition of the never-treated group, which includes firms located in Kampala, where economic activity and larger firms are disproportionately concentrated. Column (4) therefore reports statistics for never-treated firms excluding Kampala. The differences relative to ever-treated firms narrow for many variables: average revenues fall from UGX 615.4 million to UGX 552.5 million, for example, while average assets fall from UGX 271.1 million to UGX 250.7 million. Profit margins are also very similar between ever-treated firms and the non-Kampala comparison group. To address the possibility that Kampala firms exert outsized influence on the control group's trajectory, we report robustness checks throughout the analysis that estimate our specifications on the sample excluding KCC firms. These results are qualitatively and quantitatively similar to the baseline.

Importantly, our difference-in-differences design does not require treated and comparison firms to have identical outcome levels prior to district splitting. Identification

instead relies on the assumption that, absent the reform, outcomes for firms in split and unaffected districts would have evolved similarly. We therefore assess the identifying assumption using the pre-treatment dynamics reported in the event-study specifications rather than interpreting differences in baseline levels as evidence for or against the research design. As shown in the results below, the main outcomes exhibit no evidence of differential pre-trends prior to district splitting.

Table 1: Summary Statistics: Firm-Level Variables

	(1) Universe	(2) Ever-Treated	(3) Never-Treated	(4) Never-Treated (Excl. KCC)
Total Assets	262.35	182.88	271.14	250.68
	(581.79)	(442.94)	(594.53)	(559.24)
Revenues	596.57	430.88	615.44	552.46
	(1,320.95)	(1,060.77)	(1,346.14)	(1,281.90)
Profit Margins	0.14	0.18	0.14	0.17
	(0.31)	(0.30)	(0.31)	(0.32)
$\mathbb{I}\{$Loss Making$\}$	0.20	0.13	0.20	0.17
	(0.40)	(0.34)	(0.40)	(0.37)
Cost of Sales	438.34	332.57	450.39	415.45
	(1,032.93)	(877.46)	(1,048.53)	(1,026.55)
Total Employee Compensation	27.86	17.35	29.06	24.66
	(59.20)	(42.09)	(60.73)	(54.74)
Formal Wage Bill	102.51	58.42	106.54	89.01
	(196.15)	(125.22)	(200.91)	(173.48)
Formal Employees	5.74	5.19	5.80	6.22
	(11.33)	(10.89)	(11.38)	(12.14)
CIT Remitted	0.97	0.43	1.04	0.77
	(3.01)	(1.85)	(3.11)	(2.61)
PAYE Withheld	20.01	8.15	21.09	14.54
	(44.88)	(25.04)	(46.12)	(37.17)

Notes: The table reports pre-reform (2015/16) means and standard deviations for the main firm-level variables used in the analysis, with standard deviations reported in parentheses. Column 1 presents statistics for the full universe of firms in the CIT register. Columns 2 and 3 divide this universe into ever-treated and never-treated firms according to their subsequent exposure to district splitting. Ever-treated firms are located in districts that were split during the study period, regardless of the timing of the split, while never-treated firms are located in districts that remained unaffected throughout the study period. Column 4 reports statistics for never-treated firms excluding those located in the Kampala Capital City Authority (KCCA), which serves as an alternative comparison group in robustness analyses. Profit margin is defined as earnings before interest and taxes (EBIT) divided by revenues. Formal employment is measured as the number of workers on the firm's formal payroll. All monetary variables are denominated in Uganda Shillings (UGX) and expressed in constant 2016/17 prices.

IV. IDENTIFICATION STRATEGY

Identifying the causal effect of decentralization on firm performance is challenging. The decision to split a district reflects a combination of factors, including population size, geographic extent, and political considerations, some of which may be correlated with the pre-existing local economic conditions. A naive comparison of firms in split versus non-split districts would therefore confound the effects of decentralization with these underlying differences.

To identify the causal effect of decentralization on the local economy, including firm performance, we use a difference-in-differences design that exploits two sources of variation: geographic variation in exposure to the reform, since only a subset of districts were split, and variation over time in the implementation of the policy. Treatment is defined at the district level. For firm-level analysis, we assign treatment status using the firm's district of operation from the tax records. Firms located in districts that were split during the 2015/16 wave are classified as treated, while those in non-split districts form the comparison group.

A natural starting point would be to estimate the dynamic effects of the reform using a standard two-way fixed effects (TWFE) event-study regression of the form:

$$y_{it} = \phi_i + \alpha_t + \sum_{\substack{e=-5 \\ e\neq -1}}^{4} \beta_e \mathbf{1}\{t - G_i = e\} + X'_{i0}\gamma_t + \varepsilon_{it}, \tag{1}$$

where y_{it} is the outcome variable (e.g., firm performance measures) reported by firm i at time t. ϕ_i is firm fixed effect while α_t is year fixed effect. $\mathbf{1}\{t - G_i = e\}$ is an indicator for unit i being e periods away from initial treatment at time t, and G_i denotes the period when unit first becomes treated. X_{i0} is a vector of pre-treatment, time-invariant firm and location characteristics, including sector of operation, accounting month, and indicators for exposure to major place-based government programs such as the Uganda Support to Municipal Infrastructure Development (USMID) program. The term $X'_{i0}\gamma_t$ allows these characteristics to have year-specific effects, thereby allowing for differential trends. ε_{it} is an error term.[4]

However, as shown by recent literature, the TWFE estimator is potentially biased under staggered adoption due to the negative weighting induced by forbidden comparisons, whereby already-treated units serve as controls for later-treated units (Borusyak

[4]Note that equation 1 is specified at the firm level, which constitutes the core of the analysis in this paper, but an analogous specification can be estimated at the district level.

and Jaravel 2018). This bias is particularly pronounced in the presence of heterogeneous treatment effects across cohorts or over time (De Chaisemartin and d'Haultfoeuille 2020; Sun and Abraham 2021; Goodman-Bacon 2021; Callaway and Sant'Anna 2021; Borusyak et al. 2024).

To address this concern, we estimate treatment effects using the doubly robust estimator proposed by Callaway and Sant'Anna (2021), which avoids the problematic comparisons embedded in the TWFE estimator by constructing cohort-specific treatment effects using only clean comparison units. We use not-yet-treated units as the comparison group in the baseline specification, and show that all results are robust to using never-treated units instead. The estimator first recovers group-time average treatment effects, $ATT(g,t) = \mathbb{E}[Y_t(g) - Y_t(0) \mid X, G_g = 1]$, which capture the average effect of the reform on units first treated in period g, evaluated at time t. These are then aggregated into event-study parameters that summarize treatment effect dynamics by exposure length:

$$\beta_{es}(e) = \sum_{g \in \mathcal{G}} \mathbb{I}\{g + e \leq \mathcal{T}\}\, \mathbb{P}(G = g \mid G + e \leq \mathcal{T})\, ATT(g, g + e), \tag{2}$$

where e denotes event time, the number of periods since initial treatment, and G denotes the first period a unit is treated. The parameter $\beta_{es}(e)$ therefore captures the average effect of participating in the treatment e periods after adoption, averaging across all cohorts observed at event time e. This aggregation highlights treatment effect heterogeneity by exposure length while avoiding the biases associated with the TWFE approach.

The key identifying assumption is that, in the absence of district splitting, average outcomes in districts that were split would have followed the same trajectory as those in non-split districts. This parallel trends assumption does not require that the two groups are identical in levels. For instance firms in treated and comparison groups may differ in average size, sector composition, or baseline performance, but rather that their outcomes would have evolved similarly over time had the reform not occurred. We provide support for this assumption using event-study estimates to show that the coefficients in the pre-reform periods are close to zero and statistically insignificant.

While the identification strategy is presented in the context of the firm-level analysis, the same approach applies analogously to outcomes measured at the district level such as intergovernmental transfers.

V. EMPIRICAL RESULTS

V.A. Intergovernmental Grants

We begin by examining how district creation affects the fiscal resources allocated to affected local governments. This is a natural starting point for the empirical analysis because many of the potential economic effects of decentralization operate through changes in local governments' capacity to provide public goods and services. Because a newly created district does not exist as a separate unit prior to the split, we aggregate grant allocations to the parent and newly created child district after the split and treat them as a single unit throughout the sample period, thereby holding the geographic unit of analysis fixed over time. Under this construction, a positive effect implies that district creation increases the total resources allocated to the territory of the pre-split district, rather than merely dividing an existing allocation between two administrative units.

Figure 4 presents the results. Panel A shows that district creation leads to a substantial increase in total intergovernmental grant allocations per capita. The average post-treatment coefficient is 0.109 (S.E. = 0.024), implying district splitting increases allocated grants by approximately 11 percent on average. The event-study estimates show no evidence of differential pre-trends: the coefficients in the years preceding district creation are small and statistically indistinguishable from zero. Grant allocations increase immediately following the split and remain elevated throughout the post-treatment period. The persistence of the effect suggests that the increase is not limited to transitional funding associated with establishing the new district administration, but instead reflects a sustained increase in the resources allocated to the affected area.

Panels B and C decompose the increase in grant allocations into development and recurrent grants. The response is considerably larger for development grants than recurrent grants. District creation increases development grant allocations by an average of 0.227 log points (S.E. = 0.063), or approximately 23 percent. The effect is immediate and largest in the year of district creation; the point estimate declines in the following year before increasing steadily thereafter. This pattern is suggestive of an initial increase in resources associated with establishing the new district, followed by a gradual expansion in development spending as the new administrative structure becomes operational. Recurrent grants also increase, but by a smaller magnitude, with an average effect of 0.077 log points (S.E. = 0.022), approximately 7.7 percent. The increase in total grant allocations therefore does not appear to reflect recurrent expenditures alone. A substantial component of the additional resources is provided through development grants, which finance investment and other development activities undertaken by local governments.

This distinction is important for interpreting the subsequent effects on local economic activity. If district creation merely generated additional administrative overhead, an increase in grant allocations would not necessarily imply greater provision of economically productive local public goods. The relatively stronger response of development grants instead suggests that district splitting expands the resources available for investments that may affect the local business environment.

Overall, the results establish that district creation is accompanied by a meaningful expansion in central government resources allocated to the affected territory. Because the analysis aggregates the parent and child districts after a split, this increase cannot be explained mechanically by the creation of an additional budgetary unit. Rather, the combined jurisdiction receives more resources than the original district would otherwise have received. This finding is consistent with a broader literature showing that decentralization and government fragmentation can reshape the level and spatial allocation of public resources and, in turn, the provision of local public goods (Faguet 2004; Grossman et al. 2017). More directly, it parallels recent evidence from Brazil that government splitting increases local fiscal resources, which contribute to subsequent improvements in public service delivery and economic activity (Dahis and Szerman 2026). The results therefore provide an important link between the institutional reform and the local economic activity examined next using firm-level data.

One potential concern is that the increase in intergovernmental allocations to split districts may reflect a reallocation of central government resources away from districts that remained undivided, rather than an expansion in the overall transfer envelope. If this were the case, we would expect allocations to unsplit districts to stagnate or decline following the 2015/16 wave of district creation. Appendix Figure A.17 shows no evidence of such a pattern. Allocations to unsplit districts continue to increase throughout the post-reform period, both in total (Panel A) and for development (Panel B) and recurrent grants (Panel C). Allocations to split districts also continue to rise, with the gap between the two groups widening modestly in some categories after 2015. These raw trends are therefore difficult to reconcile with a simple zero-sum reallocation of resources from unsplit to split districts. Instead, they are more consistent with the additional resources received by newly created districts being accommodated through an expansion in the intergovernmental transfer envelope.

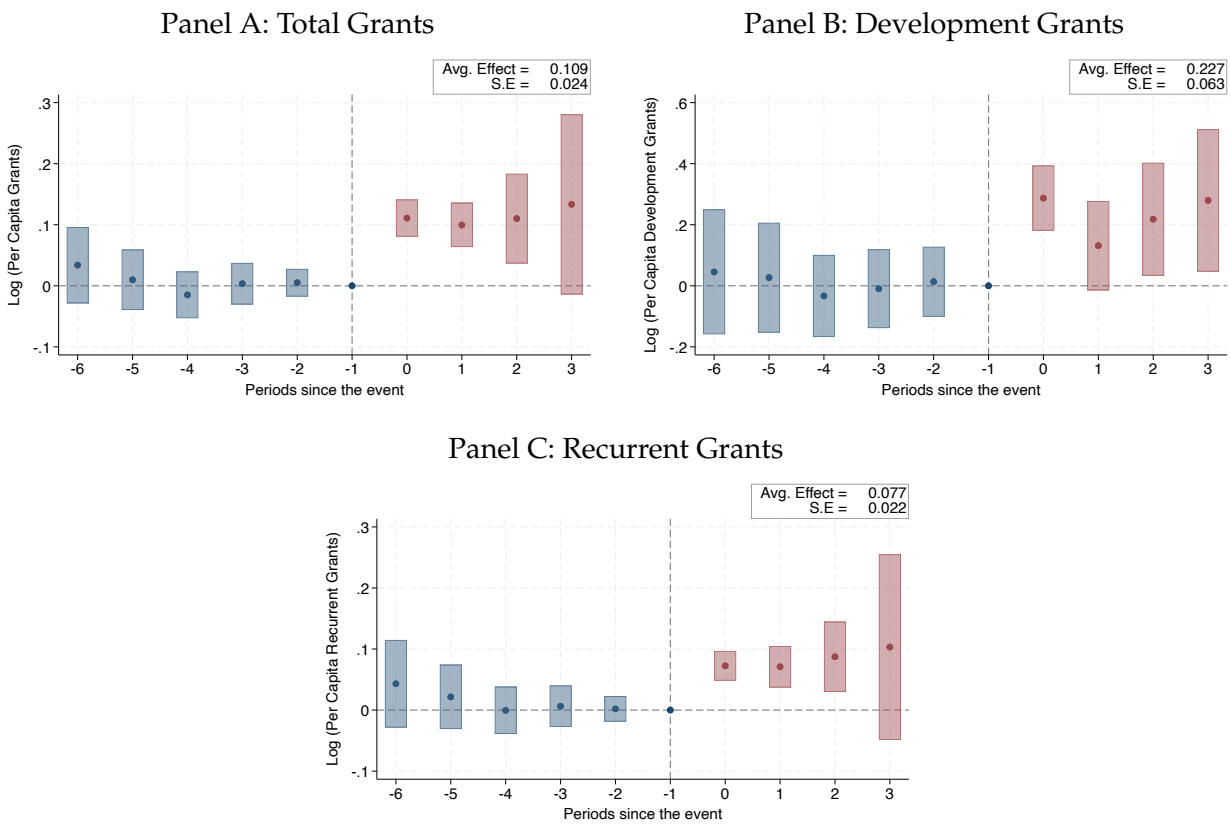


Figure 4: Effects on Intergovernmental Grant Allocation

Notes: The figure presents event-study estimates of the effects of district splitting on intergovernmental grant allocations. The unit of analysis is the pre-split district. For districts that split, grant allocations to the parent and newly created child district are aggregated after the split and treated as a single unit throughout the sample period, thereby holding the geographic unit of analysis fixed over time. The regressions are estimated using a balanced panel of 111 districts observed over eight fiscal years, yielding 888 district-year observations. Panel A reports total grant allocations per capita, Panel B reports development grant allocations, and Panel C reports recurrent grant allocations. Outcomes are expressed in logs. The horizontal axis denotes years relative to district splitting. Each dot represents the estimated treatment effect for the corresponding event year, with event time −1 omitted as the reference period; vertical bands denote 95 percent confidence intervals. The boxes report the average post-treatment effect and its standard error. Estimates are obtained using the staggered difference-in-differences estimator of Callaway and Sant'Anna (2021), with not-yet-treated districts serving as the comparison group. Standard errors are clustered at the district level.

V.B. Firm-Level Analysis

V.B.1. Firm Revenues and Profits

Having established that district creation increases the fiscal resources allocated to local governments, we next turn to firm-level outcomes, focusing first on revenues and profits, our primary measures of firm performance. While it is common in economic research to transform variables using logs, doing so in our case presents a challenge since

some of the outcomes contain zero values and negatives (in the case of profits) for some firms in certain years. In such a case, log-like transformations would present challenges in interpreting the coefficients as percentage effects as shown by Chen and Roth (2024). To circumvent this challenge, we normalize the revenues relative to year 2016 (the year before the first group of firms gets treated). This is a standard approach in literature (e.g Saez et al. (2019), Harju et al. (2022), Pinto et al. (2024)), which allows for economic interpretation of the regression coefficients, i.e in percents or percentage points. Specifically, for a given firm i we normalize the revenues as $y_{i,t}/y_{i,2016}$. Since the mean of the normalized outcome is 1 in 2016 by default, we can interpret the resulting coefficients in percents. This approach however is not suitable for profits since negative values in 2016 can alter the meaning. Instead, we normalize profits by turnover in 2016.

Figure 5 presents the dynamic effects of the district splitting on firms' turnover and gross profits. The coefficients are estimated using staggered Difference-in-Difference following the approach proposed by Callaway and Sant'Anna (2021), using the *not-yet* as controls. Appendix Figure A.11 shows that the results are similar when using never treated firms as the control. All specifications control for sector-year and accounting month fixed effects to allow for differential trends along these dimensions.

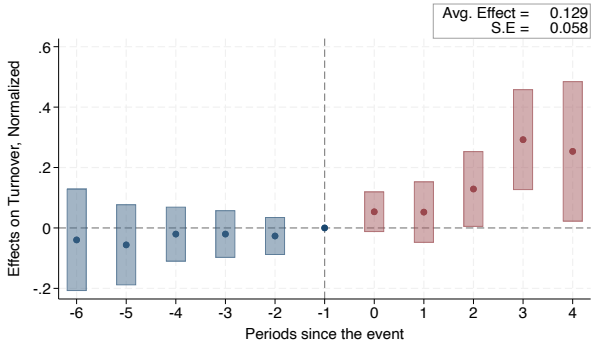
Panel A shows the effects on revenues, scaled by pre-reform values. The pre-reform coefficients are close to zero and statistically insignificant, supporting the parallel trends assumption. Following district splitting, revenues begin to increase, with the effects becoming significant approximately two years after the reform and growing steadily thereafter. The average treatment effect of 0.129 indicates the reform increased firm revenues by approximately 13 percent relative to pre-reform levels. Panel B examines the effects on gross profits, scaled by pre-reform revenues to capture changes in profit margins rather than profit levels. Despite the significant increase in revenues, the effects on profitability are small in magnitude and statistically insignificant, with an average treatment effect of 1.0 percentage points.

V.B.2. Cost of Sales and Raw Materials

The absence of significant profit effects despite a substantial increase in revenues suggests that firms' costs rose commensurately with their expanded activity. Figure 6 investigates this directly by examining the effects on cost of sales and raw material expenditures, both scaled by pre-reform revenues.[5] Panel A shows that the cost of sales increased by approximately 7.7 percentage points following district splitting, with the dynamic pattern closely mirroring that of revenues— effects emerge roughly two years after the reform

[5] Appendix Figure **??** shows a similar pattern with the outcomes scaled by their pre-reform values.

Panel A: Revenues



Panel B: Profits Before Tax/Pre-reform Revenues

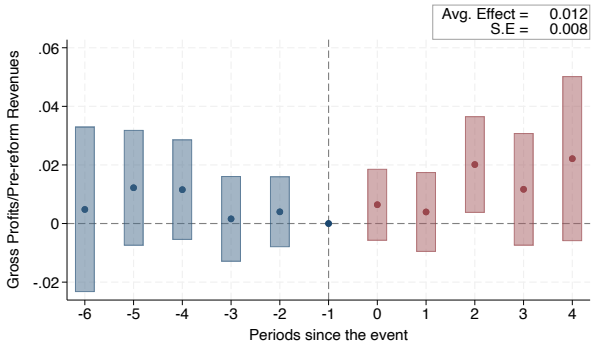


Figure 5: Effects of Firms' Revenues and Profits

Notes: The figure displays the dynamic effects of district splitting on firm turnover and profits. Turnover is normalized by its pre-reform value, while profits are normalized by pre-reform revenues. Coefficients in Panel A should therefore be interpreted as percentage changes, while those in Panels B and C should be interpreted as percentage point changes relative to pre-reform revenues. Regressions are estimated on a balanced panel of firms and weighted by the logarithm of the pre-reform revenues. The sample comprises 68,742 firm-year observations. The horizontal axis measures years relative to district splitting. Each point represents the estimated treatment effect in that year, with year -1 as the reference category. Vertical bands indicate 90 percent confidence intervals. Coefficients are estimated using a staggered difference-in-differences approach following Callaway and Sant'Anna (2021), with not-yet-treated units as the comparison group. The specification includes fixed effects for sector-year and accounting month. Standard errors are clustered at the firm level.

and grow over subsequent periods. Panel B reveals a similar pattern for raw material expenditures, which increased by approximately 6.3 percentage points. These results show that the revenue gains documented in Figure 5 were accompanied by an increases in intermediate-input expenditures, partly accounting for the limited response of profits.

V.B.3. Employee Compensation and Formal Employment

The expansion in firm activity also extends to the labor margin. Figure 7 examines total employee compensation reported on CIT returns, as well as formal wage payments and employment measured from PAYE records. Panel A shows that total employee compensation increases by an average of UGX 1.99 million (S.E. 1.05) following district splitting, equivalent to approximately 4.7 percent of the pre-reform mean of UGX 42.61 million. The dynamic estimates show no evidence of differential pre-trends and become increasingly positive after the reform, with statistically significant effects emerging in the later post-reform years.

Panels B and C show that this increase in labor expenditures is accompanied by an expansion in formal payroll and employment. The formal wage bill increases by an average

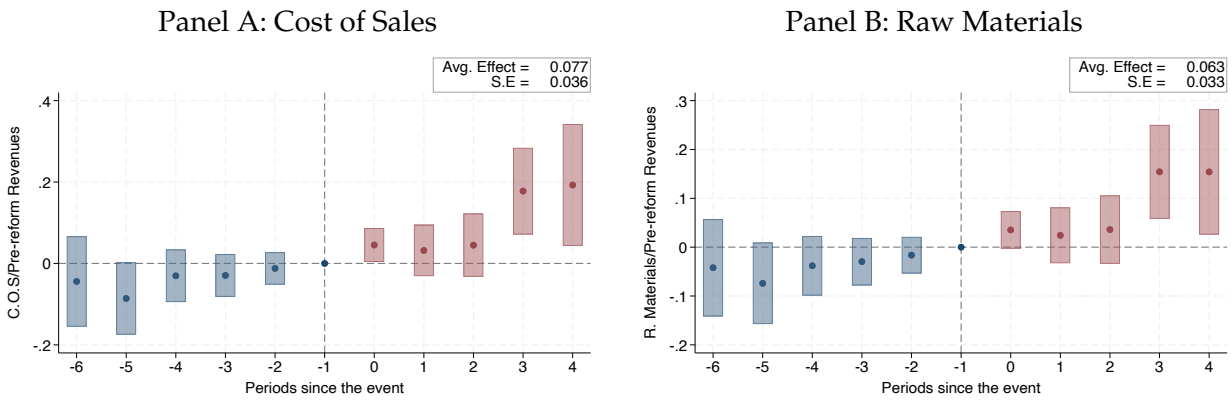


Figure 6: Cost of Sales and Raw Materials

Notes: The figure shows the dynamic effects of district splitting on firms' cost of sales and raw materials. Both outcomes are scaled by pre-reform revenues, implying the effects can be interpreted in percentage points relative to pre-reform revenues. The horizontal axis measures years relative to district splitting. Each point represents the estimated treatment effect in that year, with year -1 as the reference category. Vertical bands indicate 95 percent confidence intervals. Coefficients are estimated using a staggered difference-in-differences approach following Callaway and Sant'Anna (2021), with not-yet-treated units as the comparison group. The specification includes fixed effects for sector of operation, pre-reform firm size, and accounting month. Standard errors are clustered at the firm level.

of UGX 4.17 million (S.E. 1.68), or approximately 10.9 percent relative to its pre-reform mean of UGX 38.24 million. Formal employment also increases, by an average of 0.58 workers relative to a pre-reform mean of 7.99 workers, although the average effect is imprecisely estimated. The dynamic estimates are more informative about the timing of these responses. The formal wage bill begins to increase shortly after district splitting and is statistically distinguishable from zero from roughly the first post-reform year onward, while the employment response emerges more gradually and becomes pronounced in the later post-reform periods. The effects on total compensation similarly become largest toward the end of the event window. The timing suggests that firms may initially accommodate the expansion in activity through greater compensation of their existing formal workforce, with employment adjusting more gradually as the increase in activity persists.

These results reinforce the pattern documented for cost of sales and raw materials. The increase in revenues following district splitting is accompanied by higher expenditures on intermediate inputs and labor, as well as an expansion in formal employment. The joint response across these margins is consistent with firms scaling up productive activity rather than the revenue gains reflecting higher output prices alone. The accompanying rise in operating costs also helps explain why the substantial increase in revenues translates into a limited response of profits.

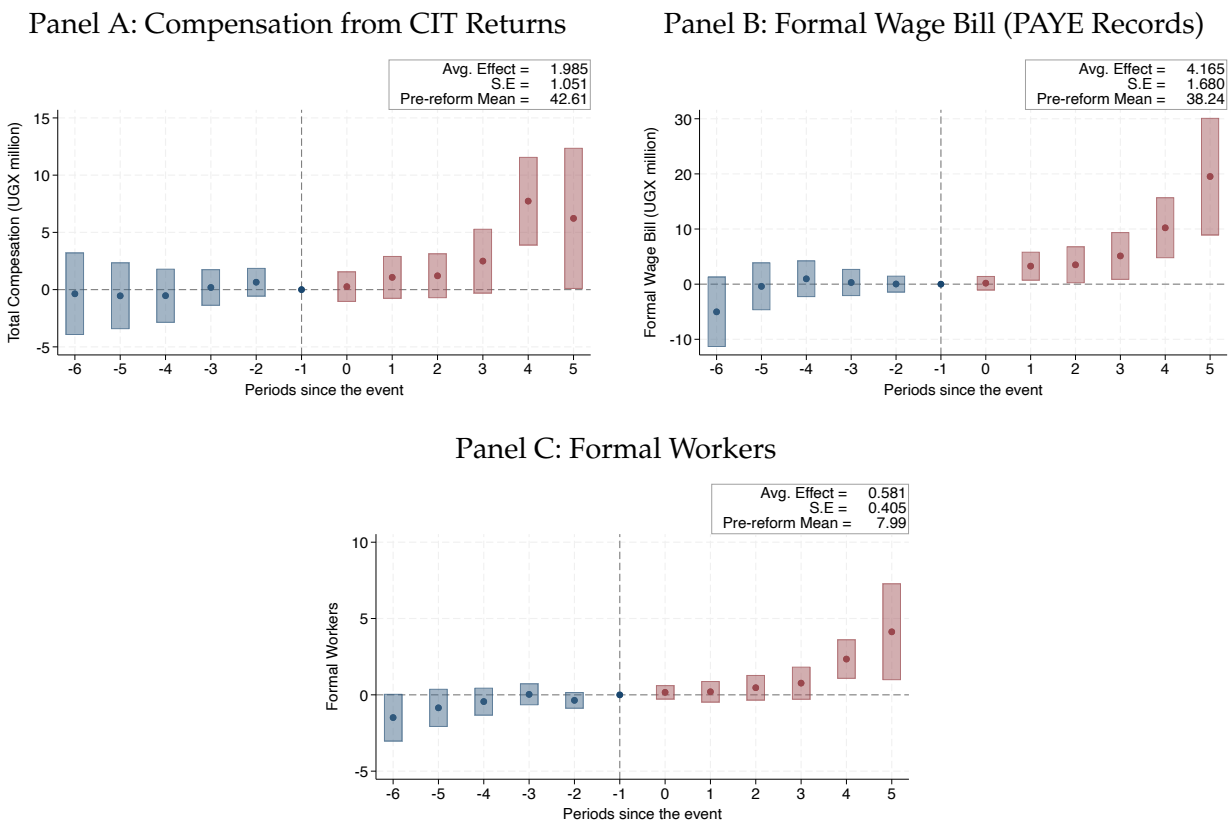


Figure 7: Effects on Employee Compensation and Formal Employment

Notes: The figure presents the effects of district splitting on firms' employee compensation and formal employment. The outcomes in Panels A and B are measured in local currency and expressed in constant 2016/17 prices, while Panel C reports the number of formal workers. Regressions are estimated on a balanced panel of firms and weighted by the logarithm of the pre-reform value of the corresponding outcome. The sample comprises 73,098 firm-year observations in Panel A and 46,305 firm-year observations in Panels B and C. The horizontal axis measures years relative to district splitting. Each dot denotes the point estimate of the treatment effect for a given year, with year −1 as the reference category, and the vertical bands denote the corresponding 95 percent confidence intervals. Coefficients are estimated using the staggered difference-in-differences estimator of Callaway and Sant'Anna (2021), with not-yet-treated units as the comparison group. The specification includes sector-year and accounting-month fixed effects. Standard errors are clustered at the firm level.

V.B.4. Robustness Checks

Changing Sample Restrictions and Comparison Group— The baseline estimates are obtained using the full sample. As a first robustness check, we also exclude districts that were split in the earlier 2009/10 wave of district creation. Although these splits predate the start of our sample, firms in these districts may still have been experiencing persistent or evolving effects of earlier decentralization during the study period. Their inclusion in the comparison group could therefore contaminate the counterfactual and, if the earlier

treatment effects operate in the same direction as those studied here, attenuate the baseline estimates. The results remain robust to this restriction and are, if anything, somewhat larger in magnitude; see Appendix Section I.A.

We also exclude firms located in Kampala, which was not among the districts affected by the reform. Because Kampala accounts for a substantial share of formal economic activity, dropping it materially changes both the size and composition of the comparison group. Reassuringly, the results are qualitatively unchanged when Kampala is excluded; see Appendix Section I.B.

Finally, we test whether the effects are sensitive to the choice of comparison group in the staggered difference-in-differences design. The baseline results use not-yet-treated units as the comparison group. Appendix Figure A.11 and A.13 shows that the results are robust to restricting the comparison group to never-treated units only.

Treatment of Outliers.— In the baseline estimates, we winsorize outcomes at the 95th percentile of the distribution. we test the sensitivity of the results to a more stringent treatment of outliers by winsorizing at the 97.5th percentile. Appendix Figure A.12 shows that the results are consistent with the baseline, albeit slightly larger in magnitude for revenues.

V.C. Mechanisms

V.C.1. Local Demand Channel

As discussed in Section II.B, a split creates a new center of local government, with its own district council, administration, and technical departments. Establishing and operating these institutions generates new public employment and purchases of goods and services, financed in large part by the increase in intergovernmental grant allocations documented in Section V.A. To the extent that public employees spend their incomes locally and newly created district governments procure goods and services from local suppliers, district creation raises the demand facing local firms. A positive demand shock of this kind would lead to expansion in firm activity.

To assess the role of local demand, we examine heterogeneity in the revenue response between firms in tradable and non-tradable sectors. This approach builds on a literature that exploits differences in firms' exposure to local demand to identify the effects of local demand shocks (Mian and Sufi 2014; Moretti 2010). If district splitting raises firm revenues by stimulating local demand, the effects should be more pronounced among firms in non-tradable sectors, whose customers are disproportionately local. Figure 8 shows

that the revenue response is statistically significant only among firms in non-tradable sectors (Panel B). The average effect over the sample period is 0.154 (S.E. = 0.085), implying an increase of approximately 15 percent relative to pre-reform levels. The corresponding estimate for firms in tradable sectors is similar in magnitude (0.172) but considerably less precise (S.E. = 0.250), potentially reflecting the relatively small number of tradable-sector firms in affected districts. Thus, while the results are consistent with a local demand channel, the evidence is not conclusive: we cannot reject that firms in tradable sectors experience revenue gains of a similar magnitude. The event-study estimates show no evidence of differential pre-trends in either sector. Appendix Figure A.14 shows no significant effects on profits for either tradable or non-tradable firms, consistent with the results for the full sample.

Overall, the results are consistent with district creation stimulating local demand, with revenue gains clearly evident among firms that primarily serve local markets. One caveat, however, is that the point estimate for tradable-sector firms is also positive and similar in magnitude, but imprecisely estimated. We therefore cannot reject that tradable firms experience revenue gains of a similar size. The evidence should therefore be interpreted as suggestive, rather than conclusive, support for a local demand channel.

V.C.2. Improved Provision of Public Goods?

A central argument for decentralization is that it improves efficiency and responsiveness in the provision of public goods and services (Bardhan 2002). Recent empirical work by Narasimhan and Weaver (2024) supports this view, showing that individuals allocated to local governments with smaller populations have better access to public goods in both the short and long run. In this subsection, we examine whether district splitting translates into improvements in public goods that are directly relevant to firm performance. we focus on electricity, which represents one of the most significant constraints on business activity in the region. According to World Bank Enterprise Survey data, 11 percent of firms in Sub-Saharan Africa cite electricity as the single biggest obstacle to doing business, the highest share of any region. Firms and households routinely endure extended outages, and even when electricity is nominally available, supply is often unreliable. Given data limitations, we focus specifically on the reliability of electricity supply rather than access or cost.

To examine how district splitting affects the reliability of electricity, we draw on household panel data from the Uganda National Panel Surveys, collected under the Living Standards Measurement Study (LSMS) by the World Bank. The relevant survey question asks respondents how many hours per day they typically have power. Although

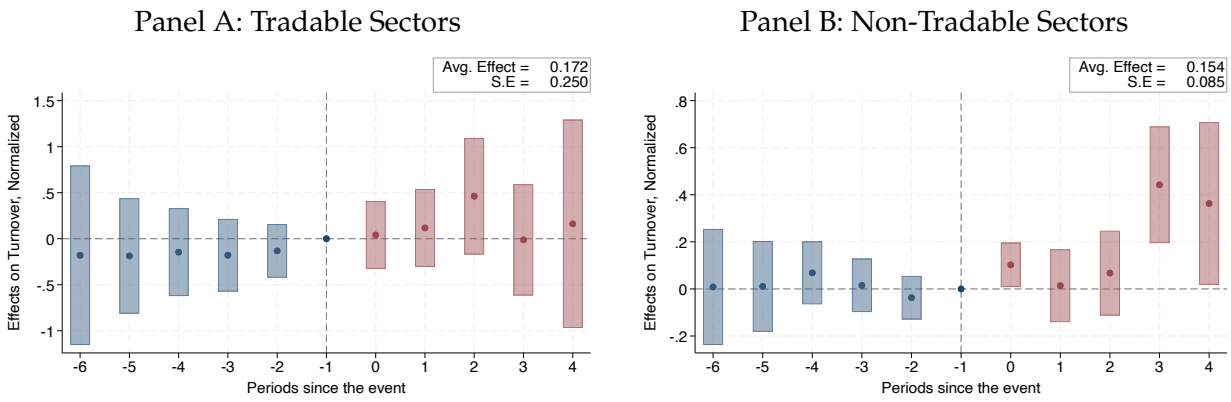


Figure 8: Effects on Firm Revenues: Tradable vs Non-tradable

Notes: The figure presents the effects of the reform on firm revenues separately for firms in tradable and non-tradable sectors. Tradable sectors include agriculture, mining and quarrying, manufacturing, and electricity and gas production. Non-tradable sectors include water supply and sewerage management, construction, retail trade, transport and storage, accommodation and food services, real estate, professional services, and administrative and support services. The classification excludes sectors that cannot be cleanly categorized as either tradable or non-tradable, such as wholesale trade. The sample comprises 7,110 firm-year observations in Panel A and 26,163 firm-year observations in Panels B. The horizontal axis measures years relative to district splitting. Each dot denotes the point estimate of the treatment effect for a given year, with year −1 as the reference category, and the vertical bands denote the corresponding 95 percent confidence intervals. Coefficients are estimated using the staggered difference-in-differences estimator of Callaway and Sant'Anna (2021), with not-yet-treated units as the comparison group. The estimations condition on the sector of operation and the pre-reform accounting. Standard errors are clustered at the firm level.

this question is administered to households rather than firms, it provides an informative measure of the general reliability of electricity supply in a given location. Since firms and households within the same area are typically served by the same distribution infrastructure, improvements in household electricity reliability are likely to reflect corresponding improvements for local businesses, making this a useful proxy for examining the role of electricity as a channel through which decentralization affects firm performance.

Results in Table 2 show that decentralization has a positive and significant effect on the reliability of electricity supply. Following district splitting, the number of hours that power is available increases by 13 percent. The effect is larger when Kampala is excluded from the sample, suggesting that the baseline estimate is conservative. This finding is consistent with newly created districts having greater lobbying power and political visibility, enabling them to attract improvements in critical public infrastructure such as electricity provision.

Table 2: Effects of Decentralization on Reliability of Electricity

	Full Sample		Excluding the Capital Ciy	
	(1)	(2)	(3)	(4)
Split x Post	0.124**	0.125**	0.171***	0.175***
	(0.063)	(0.063)	(0.056)	(0.056)
District FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Urban x Year FE	No	Yes	No	Yes
Mean at the Baseline (Hours)	19.1	19.1	19.1	19.1
Observations	1965	1965	1390	1390

Note: The table presents the average effects of district splitting on reliability of electricity. The dependent variable is hours in a typical day that power is on. Column 1 and 2 show the effects in the full sample while column 3 and 4 show the effects while excluding the capital from the sample. All coefficients are estimated using Poisson Pseudo Maximum Likelihood (PPML) and therefore should be interpreted in percents. Standard errors are clustered at district level. *** p<0.01, ** p< 0.05, * p<0.10.

V.D. Heterogeneity: Evenness of the Splits

The effects of decentralization may depend not only on whether a district is split but on how it is split. When a parent district is subdivided, the resulting fragments may be roughly equal in size or highly asymmetric, with a dominant rump retaining the vast majority of the population and economic activity. Such asymmetry can arise when district creation is motivated less by administrative efficiency than by political considerations, for example, decentralization being pursued to diffuse social and political tensions or to grant cultural and political autonomy to specific subgroups within a jurisdiction (Bardhan 2002), in which case the new district may be carved along ethnic or political lines rather than along boundaries that would produce fiscally viable jurisdictions. The balance of the split therefore has direct implications for the administrative and fiscal capacity of the resulting districts. An even split produces jurisdictions of comparable size, each with a sufficient tax base and population to sustain local public goods provision. An uneven split, by contrast, may leave the new district too small to achieve the minimum efficient scale for service delivery, while the rump parent retains enough resources that its governance structure is largely undisrupted.

To assess these considerations empirically, we first examine how evenly the districts in our study were split, using pre-reform population data to compute the resulting population shares of the parent and its newly created fragments. For a simple two-way split, the share of population allocated to each fragment provides an intuitive and sufficient measure of balance; a 50-50 split is perfectly even, while a 90-10 split is highly asymmetric.

However, this approach does not generalize to cases where a parent district is subdivided into more than two fragments, as is the case for some districts in my sample, since there is no natural way to reduce three or more population shares to a single measure of balance. we therefore construct a formal evenness index that accommodates splits of any size and is comparable across two-way and three-way divisions.

The index is derived from the Herfindahl-Hirschman Index (HHI) computed over the population shares of the resulting fragments. For a parent district that is divided into n fragments (including the rump parent), let s_i denote the population share of fragment i i, where $\sum_{i=1}^{n} s_i = 1$. The HHI is defined as:

$$HHI = \sum_{i=1}^{n} s_i^2 \tag{3}$$

The HHI ranges from $1/n$ (when all fragments are of equal size) to 1 (when the entire population is concentrated in a single fragment). Since the lower bound of this value depends on n, raw values are not directly comparable across splits involving different numbers of fragments. To address this, we normalize the HHI as:

$$HHI^{norm} = \frac{HHI - 1/n}{1 - 1/n} \tag{4}$$

The normalized HHI equals zero when all fragments are of equal size and approaches one under maximum concentration. The evenness index is then defined as:

$$E = 1 - HHI^{norm} \tag{5}$$

This index is bounded between 0 and 1, where $E = 1$ indicates a perfectly even split and $E = 0$ indicates maximum asymmetry. Importantly, the normalization ensures that the index is comparable across splits involving different numbers of fragments: a two-way split producing equal halves and a three-way split producing equal thirds both receive an evenness score of 1. Figure 9 presents the index constructed following this approach.

We then examine whether the effects of decentralization vary with the balance of the split. We divide the sample at the median of the evenness index and estimate the effects separately for firms located in districts with below-median and above-median evenness scores. Figure 10 presents the results. The effects on firm revenues are concentrated entirely among firms in districts that were split more evenly (Panel B), where revenues increase by approximately 15 percent following district splitting. By contrast, firms in districts with below-median evenness scores show no significant effects (Panel A). This pattern is consistent with the view that the benefits of decentralization depend on whether

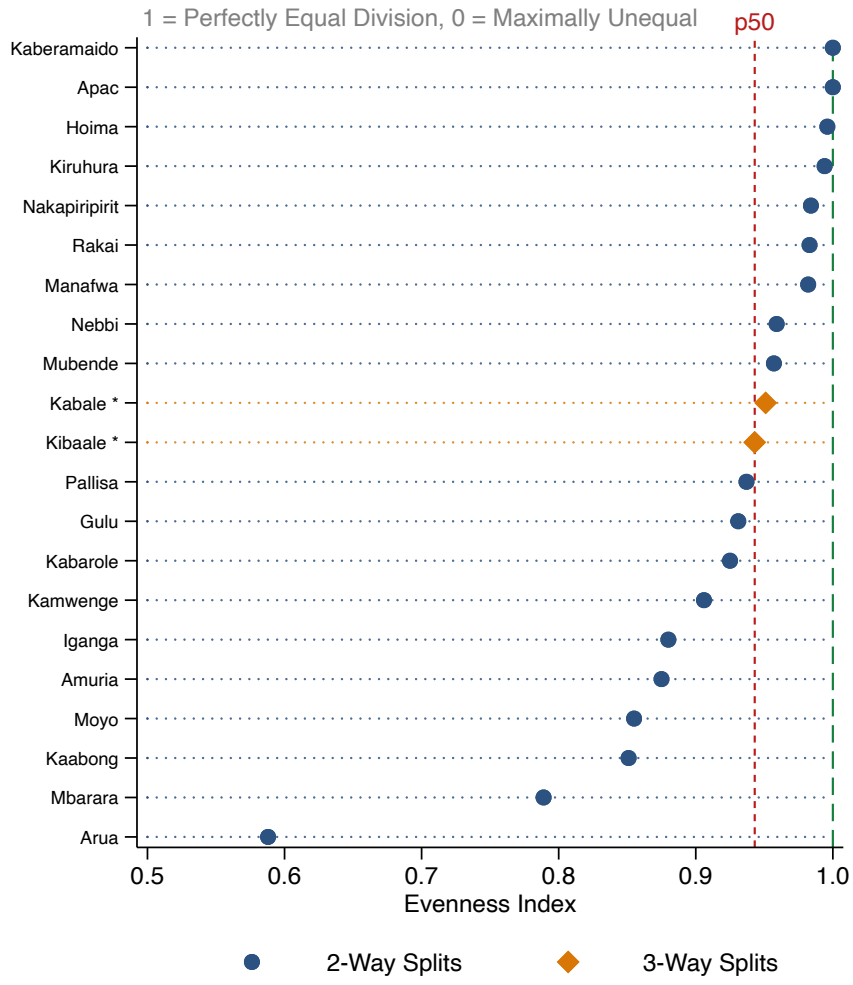


Figure 9: Evenness Index for Split Districts

Notes: The figure displays the evenness index for each parent district that was split during the 2015/16 wave of district creation, constructed as described in Section V.D. Each dot represents a single parent district. The horizontal axis measures the evenness index, which ranges from 0 (maximally asymmetric split) to 1 (perfectly equal split). The vertical dashed line indicates the median evenness score across all splits.

the resulting jurisdictions are of sufficient size to sustain effective administration and public goods delivery. Highly asymmetric splits, which may produce fragments too small to operate effectively, do not generate the same gains for local firms.

V.E. State Capacity and Revenue Effects

A central debate in the decentralization literature is whether devolving authority strengthens or weakens the state's fiscal capacity (Bardhan and Mookherjee 2000; Faguet 2014; Besley and Persson 2009). On the one hand, decentralization may erode revenue mobilization if newly created subnational units lack the administrative capacity to enforce

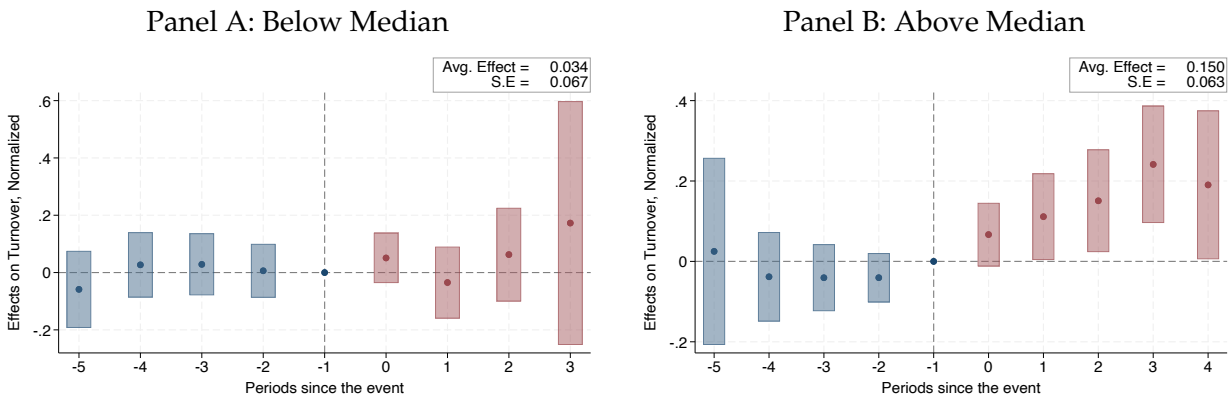


Figure 10: Heterogeneity Based on Evenness of the Splits

Notes: The figure presents the dynamic effects of decentralization on firm revenues, estimated separately for firms in districts with below-median (Panel A) and above-median (Panel B) evenness scores. The evenness index measures how balanced the population split was when the parent district was subdivided, as described in Section V.D. The dependent variable is firm turnover scaled by its pre-reform value, so that coefficients can be interpreted as percent changes. Each dot denotes the point estimate of the treatment effect for a given year, with year −1 as the reference category. The vertical bands denote the corresponding 95 percent confidence intervals. All specifications include fixed effects for sector, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

compliance or engage in tax competition. On the other hand, bringing government closer to taxpayers may enhance state legitimacy, reduce the informational costs of enforcement, and strengthen the fiscal contract (Besley 2020; Weigel 2020). In our setting, corporate income tax is collected by the central government, allowing us to examine whether decentralization generates fiscal-capacity gains that extend beyond the revenues of subnational governments.

To examine this question, we focus on two outcomes available in the administrative tax data: corporate income tax (CIT) remitted and the effective tax rate (ETR), defined as tax paid divided by pre-tax profits and set to missing for firm-years with negative profits. Applying the identification strategy described in Section IV, we find an increase in both CIT payments and the effective tax rate among firms in split districts relative to those in unaffected districts following the reform. Figure 11, Panel A, shows that annual CIT payments increase by an average of UGX 0.252 million, equivalent to approximately 23 percent of the pre-reform mean of UGX 1.11 million. Panel B shows that the effective tax rate increases by 1.99 percentage points relative to a pre-reform mean of 6.42 percent, an increase of approximately 31 percent. For both outcomes, the dynamic estimates show no evidence of differential pre-trends, supporting the parallel-trends assumption. Because the analysis is based on a balanced panel, the increase in CIT payments reflects higher remittances among continuing firms rather than changes in firm composition arising from

entry or exit. Estimates from the unbalanced panel, reported in Appendix Figure A.15, are qualitatively similar. Further robustness checks that exclude Kampala from the sample (Appendix Figure A.9) and exclude districts split in 2009 and 2010 (Appendix Figure A.4) also produce similar results.

An increase in CIT payments could arise mechanically if decentralization raises firm profitability and thereby expands the corporate income tax base. The results on firm performance, however, provide limited support for this explanation: the estimated effects on pre-tax profits are positive but imprecisely estimated. The increase in the effective tax rate is therefore particularly informative because it captures tax payments relative to reported pre-tax profits. If the increase in CIT revenue were driven solely by higher profitability, with the effective degree of taxation unchanged, tax payments and profits would be expected to increase proportionately, leaving the effective tax rate approximately unchanged. Instead, the effective tax rate rises alongside CIT payments, indicating that firms remit more tax conditional on their reported profit base.

Additional evidence is consistent with this interpretation. Appendix Figure A.16 examines deferred tax liabilities, scaled by pre-reform revenues. The reform reduced deferred tax by approximately 0.005 percentage points of pre-reform revenues on average, a proportional decrease of about 17 percent relative to a pre-reform mean of 0.03. A decline in the stock of deferred liabilities implies that firms are recognizing a larger share of their tax obligations in the current period rather than deferring them into the future. This provides a potential mechanism for the ETR result: current-period tax payments and the current effective tax rate rise, even when underlying profitability is unchanged, because firms are unwinding the timing strategies that previously allowed them to defer tax. Taken together, these results shift the interpretation away from a pure tax-base effect and toward an increase in effective tax collection: following decentralization, firms remit more tax relative to their reported profits while deferring less tax to future periods.

This interpretation is consistent with several potentially complementary mechanisms. By bringing public institutions closer to firms, decentralization may make the state more visible and responsive, strengthening the perceived fiscal exchange between taxation and public services and thereby increasing voluntary compliance (Bodea and LeBas 2016; Ali et al. 2014; Besley 2020). Greater local political representation may similarly reduce perceptions of exclusion or unequal treatment, which have been associated with lower tax-compliance attitudes in African settings (Ali et al. 2014). Decentralization may also improve the information available to public authorities and facilitate interactions between taxpayers and the state, increasing the government's ability to identify tax liabilities and enforce payment (Balán et al. 2022). More broadly, evidence from decentralized fiscal

systems suggests that greater local autonomy can strengthen tax morale and taxpayers' willingness to contribute to government (Güth et al. 2005; Torgler et al. 2010).

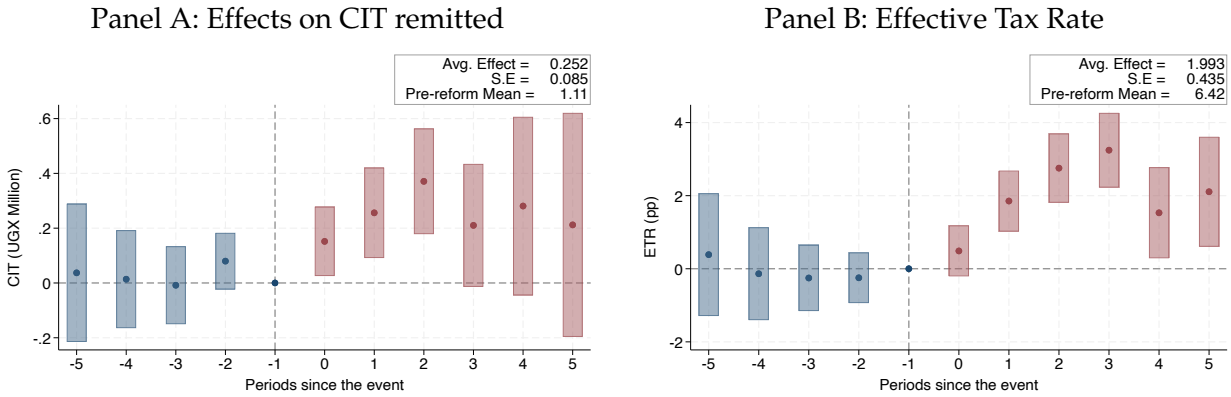


Figure 11: Effects on Corporate Income Tax Payment

Notes: The figure presents the effects of decentralization on corporate income tax remitted by firms (Panel A) and on the effective tax rate (Panel B). The effective tax rate is computed as the ratio of tax paid to pre-tax profits and is set to missing for firms with negative profits, which are thereby dropped from the sample. Regressions are estimated on a balanced panel, with observations weighted by the logarithm of pre-reform revenues. The sample comprises 80,100 firm-year observations in Panel A and 68,488 in Panel B. Each dot denotes the point estimate of the treatment effect for a given year, with year −1 as the reference category, and the vertical bands denote the corresponding 95 percent confidence intervals. All specifications include sector-year and accounting-month fixed effects. Standard errors are clustered at the firm level.

The revenue gains from decentralization extend beyond the corporate income tax. Firms also remit pay-as-you-earn (PAYE) income tax on behalf of their employees, providing a second channel through which decentralization can affect central government revenues. Figure 12 Panel A shows that annual PAYE payments rise by an average of UGX 0.29 million (S.E. 0.15) following district splitting, an increase of approximately 4.7 percent relative to the pre-reform mean of UGX 6.25 million. The dynamic estimates are statistically indistinguishable from zero prior to district splitting, and become increasingly positive after the reform, with the effects growing over time. This pattern is consistent with the increase in the formal wage bill documented in Section V.B.3: as firms expand formal labor compensation, the associated PAYE tax base expands mechanically. The PAYE response therefore appears to reflect, at least in part, growth in taxable formal labor income rather than a change in effective taxation

To assess the overall revenue implications for the central government, Panel B combines CIT and PAYE remittances. Total annual income-tax payments increase by approximately UGX 0.552 million, relative to a pre-reform mean of UGX 7.36 million, corresponding to an increase of about 7.5 percent. Thus, decentralization generates a broader increase in centrally collected tax revenue that reflects both an expansion in the formal

labor tax base and higher corporate income tax remittances.

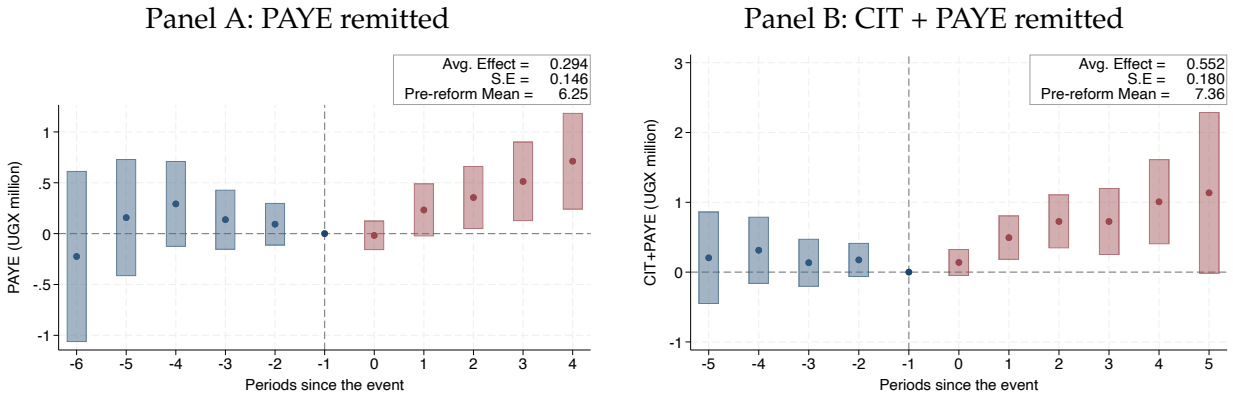


Figure 12: Effects on Tax Remitted

Notes: The figure presents the effects of decentralization on PAYE tax (Panel A) and on the total tax remitted (Panel B). Regressions are estimated on a balanced panel, with observations weighted by the logarithm of pre-reform revenues. The sample comprises 68,742 firm-year observations. Each dot denotes the point estimate of the treatment effect for a given year, with year −1 as the reference category, and the vertical bands denote the corresponding 95 percent confidence intervals. All specifications include sector-year and accounting-month fixed effects. Standard errors are clustered at the firm level.

VI. CONCLUSION

This paper examines whether decentralization generates tangible economic gains for the private sector and how it affects the fiscal capacity of the central government. Exploiting the 2015/16 wave of district creation in Uganda, we first show that district splitting substantially increases the fiscal resources allocated to the affected local governments. Total intergovernmental grant allocations per capita rise by approximately 11 percent, with a considerably larger increase in development grants than in recurrent grants. Because we aggregate allocations to the parent and newly created child districts after a split, these effects reflect a genuine increase in resources to the affected territory rather than a mechanical division of existing funds across additional administrative units.

Turning to firm outcomes, we find that decentralization increases the revenues of local firms, with effects emerging approximately two years after district splitting and growing steadily thereafter. The revenue gains are accompanied by proportional increases in expenditures on intermediate inputs and labor, as well as an expansion in formal employment and the wage bill, leaving profit margins unchanged. This pattern is consistent with firms expanding the scale of their operations in response to an improved local business environment, rather than becoming more profitable on the margin. We provide evidence

that increased local demand and improvements in local infrastructure, including greater electricity reliability, help explain the effects on firm activity.

The gains from decentralization, however, depend critically on how new administrative units are created. Revenue gains are concentrated entirely among firms in districts that were split symmetrically, while highly asymmetric splits, which may produce fragments too small to command sufficient resources and political leverage for effective public goods delivery, generate no significant effects. This heterogeneity carries a first-order policy implication: the returns to decentralization depend not only on whether administrative units are subdivided, but on how their boundaries are drawn, a design margin that has received limited attention in decentralization literature.

We also find that decentralization strengthens the fiscal capacity of the central government. Corporate income tax remittances increase alongside the effective tax rate, while profits respond only weakly, a pattern consistent with improved effective taxation rather than an expansion of the corporate tax base alone. At the same time, the increase in formal employment and the formal wage bill expands the PAYE tax base, generating additional labor income tax revenue. Thus, decentralization appears to raise central government revenue through two complementary channels: stronger effective taxation of firms and an expansion of the formal labor tax base.

Overall, the findings show that decentralization can generate benefits that extend beyond the public sector itself. By improving the local business environment, it can induce firms to expand productive activity and formal employment, while simultaneously increasing the central government's capacity to mobilize revenue. These gains are not automatic, however. They depend importantly on the scale and configuration of the newly created jurisdictions. The results therefore highlight a broader lesson for the design of decentralization reforms in low-income countries: administrative proximity can improve both economic performance and fiscal capacity, but only when decentralization preserves the institutional and economic scale necessary for effective public-good provision.

REFERENCES

Alesina, A. and Spolaore, E. (1997), 'On the number and size of nations', *The Quarterly Journal of Economics* **112**(4), 1027–1056.

Ali, M., Fjeldstad, O.-H. and Sjursen, I. H. (2014), 'To pay or not to pay? citizens' attitudes toward taxation in kenya, tanzania, uganda, and south africa', *World Development* **64**, 828–842.

Awortwi, N. (2011), 'An unbreakable path? a comparative study of decentralization and local government development trajectories in ghana and uganda', *International Review of Administrative Sciences* **77**(2), 347–377.

Balán, P., Bergeron, A., Tourek, G. and Weigel, J. L. (2022), 'Local elites as state capacity: How city chiefs use local information to increase tax compliance in the Democratic Republic of the Congo', *American Economic Review* **112**(3), 762–797.

Bardhan, P. (2002), 'Decentralization of governance and development', *Journal of Economic perspectives* **16**(4), 185–205.

Bardhan, P. and Mookherjee, D. (2000), 'Capture and governance at local and national levels', *American Economic Review* **90**(2), 135–139. Papers and Proceedings.

Besley, T. (2020), 'State capacity, reciprocity, and the social contract', *Econometrica* **88**(4), 1307–1335.

Besley, T. and Persson, T. (2009), 'The origins of state capacity: Property rights, taxation, and politics', *American Economic Review* **99**(4), 1218–1244.

Bodea, C. and LeBas, A. (2016), 'The origins of voluntary compliance: Attitudes toward taxation in urban nigeria', *British Journal of Political Science* **46**(1), 215–238.

Boffa, F., Piolatto, A. and Ponzetto, G. A. (2016), 'Political centralization and government accountability', *The Quarterly Journal of Economics* **131**(1), 381–422.

Borusyak, K. and Jaravel, X. (2018), *Revisiting event study designs*, Vol. 2826228, SSRN.

Borusyak, K., Jaravel, X. and Spiess, J. (2024), 'Revisiting event-study designs: robust and efficient estimation', *Review of Economic Studies* **91**(6), 3253–3285.

Callaway, B. and Sant'Anna, P. H. (2021), 'Difference-in-differences with multiple time periods', *Journal of econometrics* **225**(2), 200–230.

Cassidy, T. and Velayudhan, T. (2026), 'Government fragmentation and economic growth', *Review of Economics and Statistics* pp. 1–18.

Chen, J. and Roth, J. (2024), 'Logs with zeros? some problems and solutions', *The Quarterly Journal of Economics* **139**(2), 891–936.

Cohen, I. (2024), 'Documenting decentralization: Empirical evidence on administrative unit proliferation from uganda', *The World Bank Economic Review* p. lhae008.

Dahis, R. and Szerman, C. (2024), 'Decentralizing development: Evidence from government splits'.

Dahis, R. and Szerman, C. (2026), 'Decentralizing development: The economic impacts of government splits', *American Economic Journal: Applied Economics* **18**(3), 220–253.

De Chaisemartin, C. and d'Haultfoeuille, X. (2020), 'Two-way fixed effects estimators with heterogeneous treatment effects', *American Economic Review* **110**(9), 2964–2996.

Faguet, J.-P. (2004), 'Does decentralization increase government responsiveness to local needs?: Evidence from bolivia', *Journal of public economics* **88**(3-4), 867–893.

Faguet, J.-P. (2014), 'Decentralization and governance', *World Development* **53**, 2–13.

Gerring, J., Palmer, M., Teorell, J. and Zarecki, D. (2015), 'Demography and democracy: A global, district-level analysis of electoral contestation', *American Political Science Review* **109**(3), 574–591.

Goodman-Bacon, A. (2021), 'Difference-in-differences with variation in treatment timing', *Journal of econometrics* **225**(2), 254–277.

Green, E. (2010), 'Patronage, district creation, and reform in uganda', *Studies in comparative international development* **45**(1), 83–103.

Grossman, G. and Lewis, J. I. (2014), 'Administrative unit proliferation', *American Political Science Review* **108**(1), 196–217.

Grossman, G., Pierskalla, J. H. and Boswell Dean, E. (2017), 'Government fragmentation and public goods provision', *The Journal of Politics* **79**(3), 823–840.

Güth, W., Levati, M. V. and Sausgruber, R. (2005), 'Tax morale and (de-)centralization: An experimental study', *Public Choice* **125**(1–2), 171–188.

Harju, J., Koivisto, A. and Matikka, T. (2022), 'The effects of corporate taxes on small firms', *Journal of Public Economics* **212**, 104704.

Hatfield, J. W. (2015), 'Federalism, taxation, and economic growth', *Journal of urban economics* **87**, 114–125.

Lewis, B. D. (2017), 'Does local government proliferation improve public service delivery? evidence from indonesia', *Journal of Urban Affairs* **39**(8), 1047–1065.

Mian, A. and Sufi, A. (2014), 'What explains the 2007–2009 drop in employment?', *Econometrica* **82**(6), 2197–2223.

Moretti, E. (2010), 'Local multipliers', *American Economic Review* **100**(2), 373–377.

Narasimhan, V. and Weaver, J. (2024), 'Polity size and local government performance: evidence from india', *American Economic Review* **114**(11), 3385–3426.

Pinto, D. P., Bermúdez, J. C., de Arruda, T. D. G. S. et al. (2024), Vat refunds and firms' performance: Evidence from a withholding reform in honduras, Technical report, The World Bank.

Republic of Uganda (1995), 'Constitution of the republic of uganda'. As amended.
URL: *https://ulii.org/en/akn/ug/act/statute/1995/constitution/eng@2023-12-31*

Republic of Uganda (1997), 'Local governments act'. Chapter 138, as amended.
URL: *https://ulii.org/en/akn/ug/act/1997/5/eng@2023-12-31*

Saez, E., Schoefer, B. and Seim, D. (2019), 'Payroll taxes, firm behavior, and rent sharing: Evidence from a young workers' tax cut in sweden', *American Economic Review* **109**(5), 1717–1763.

Singhania, D. (2022), 'Do smaller local governments bring citizens more? evidence from direct elections in indonesia', *The World Bank Economic Review* **36**(3), 774–799.

Stansel, D. (2005), 'Local decentralization and local economic growth: A cross-sectional examination of us metropolitan areas', *Journal of Urban Economics* **57**(1), 55–72.

Sun, L. and Abraham, S. (2021), 'Estimating dynamic treatment effects in event studies with heterogeneous treatment effects', *Journal of econometrics* **225**(2), 175–199.

Torgler, B., Schneider, F. and Schaltegger, C. A. (2010), 'Local autonomy, tax morale, and the shadow economy', *Public Choice* **144**(1–2), 293–321.

Weigel, J. L. (2020), 'The participation dividend of taxation: How citizens in Congo engage more with the state when it tries to tax them', *Quarterly Journal of Economics* **135**(4), 1849–1903.

World Bank (2024), Uganda public expenditure review 2022–23: Module ii (b) – intergovernmental fiscal transfers, Technical report, World Bank, Washington, DC.
URL: *https://documents1.worldbank.org/curated/en/099041224173016094/pdf/P1771821adfe590c118a791e8*

Zhang, T., Sun, B., Cai, Y. and Wang, R. (2019), 'Government fragmentation and economic growth in china's cities', *Urban Studies* **56**(9), 1850–1864.

APPENDIX

CONTENTS

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I. ROBUSTNESS TO DIFFERENT SAMPLE RESTRICTIONS

I.A. Excluding Districts Split in 2009 and 2010 from the Sample

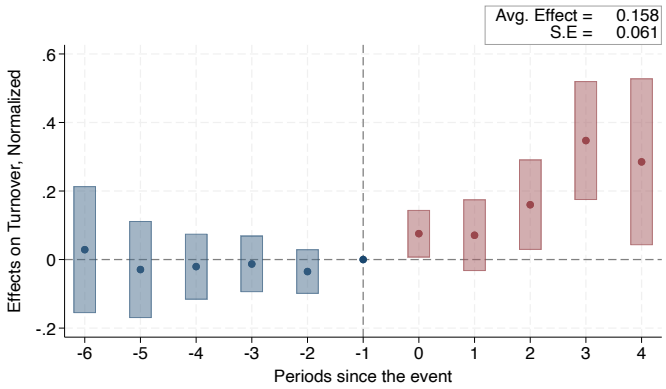
The baseline comparison group includes firms located in districts that were split in 2009 or 2010, prior to the beginning of our administrative data in 2013/14. Because these district splits predate the sample period, we do not observe firms in these locations prior to treatment and therefore classify them as untreated within the observed estimation window. This raises a potential concern if the effects of district splitting persist or continue to evolve several years after a district is created, as our main event-study estimates suggest they do.

The presence of these previously treated districts in the comparison group does not necessarily threaten identification. If the effects of the 2009 and 2010 splits had fully stabilized before 2013/14, any persistent level differences would be absorbed by the fixed effects and would not affect identification from differential changes over time. The concern instead arises if firms in these districts were still adjusting to earlier decentralization during our sample period. In that case, their outcomes would reflect ongoing treatment effects rather than the counterfactual evolution of untreated districts, contaminating the comparison group. If these continuing effects operate in the same direction as those generated by the district splits studied in our main analysis, their inclusion would tend to attenuate the estimated treatment effects. More generally, however, the direction of the resulting bias depends on the dynamics of the earlier treatment effects.

To address this concern, we re-estimate all of the main specifications after excluding districts that were split in 2009 or 2010 from the analysis sample. This restriction ensures that the comparison group does not include districts with known exposure to district splitting closer to the start of our data.

Appendix Figures A.1 through A.5 present the results for each of the main outcomes examined in the paper. The results, reported below, are qualitatively unchanged and display very similar dynamic patterns to the baseline estimates. Moreover, the estimated effects are generally somewhat larger in magnitude across the main outcomes. This pattern is consistent with the possibility that firms in districts split before the start of our sample were still experiencing positive effects of decentralization during the study period, causing their inclusion in the comparison group to attenuate the baseline estimates. The robustness of the results to excluding these districts therefore reinforces the main findings and suggests that, if anything, the baseline estimates may be conservative.

Panel A: Turnover



Panel B: Gross Profits/Pre-reform Revenues

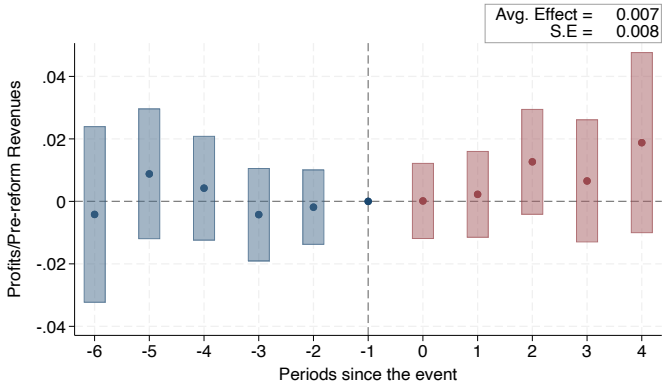
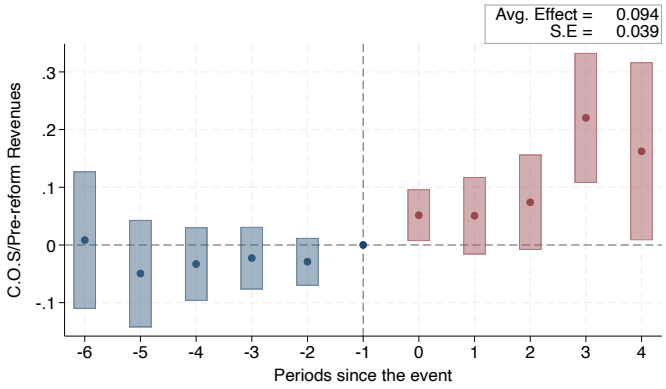


Figure A.1: Effects of Firms' Revenues and Profits: Excluding 2009/10 Splits

Notes: The figure shows the dynamic effects of district splitting on firms' turnover and profits. The comparison group is firms in split districts excluding the districts that were split in 2009/10 reform. Turnover is normalized by its pre-reform value, while profits are normalized by pre-reform revenues. Coefficients in Panel A should therefore be interpreted as percentage changes, while those in Panels B should be interpreted as percentage point changes relative to pre-reform revenues. The horizontal axis measures years relative to district splitting. Each point represents the estimated treatment effect in that year, with year -1 as the reference category. Vertical bands indicate 95 percent confidence intervals. The specification includes fixed effects for sector of operation, pre-reform firm size, and accounting month. Standard errors are clustered at the firm level.

Panel A: Cost of Sales



Panel B: Raw Materials

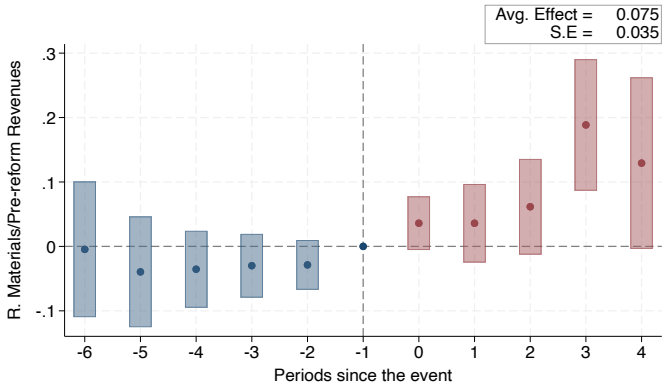


Figure A.2: Cost of Sales and Raw Materials: *Excluding 2009/10 Splits*

Notes: The figure displays the dynamic effects of district splitting on firms' cost of sales and raw materials, excluding 2009/10 splits. Each outcome is scaled by its pre-reform values, implying the coefficients can be interpreted in percent changes. The horizontal axis measures years relative to district splitting. Each point represents the estimated treatment effect in that year, with year −1 as the reference category. Vertical bands indicate 95 percent confidence intervals. Coefficients are estimated using a staggered difference-in-differences approach following Callaway and Sant'Anna (2021), with not-yet-treated units as the comparison group. The specification includes fixed effects for sector of operation, pre-reform firm size, and accounting month. Standard errors are clustered at the firm level.

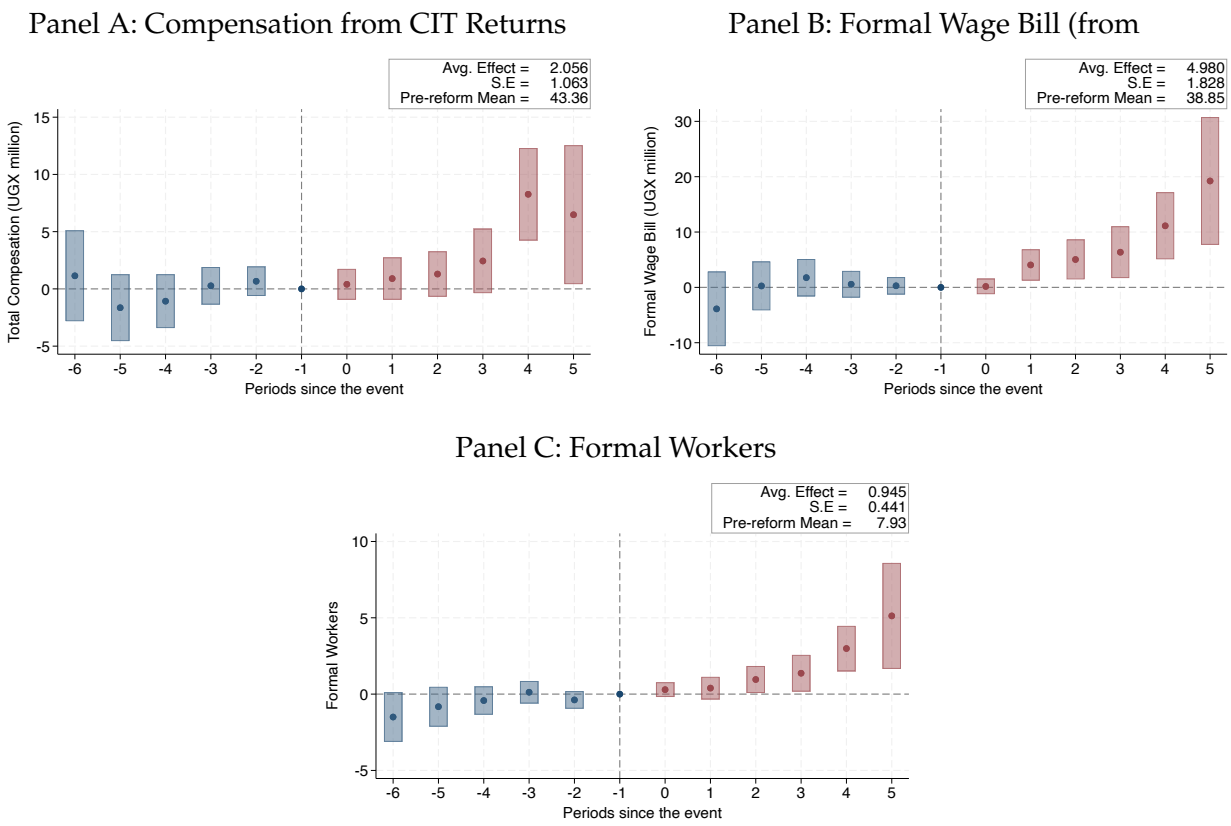
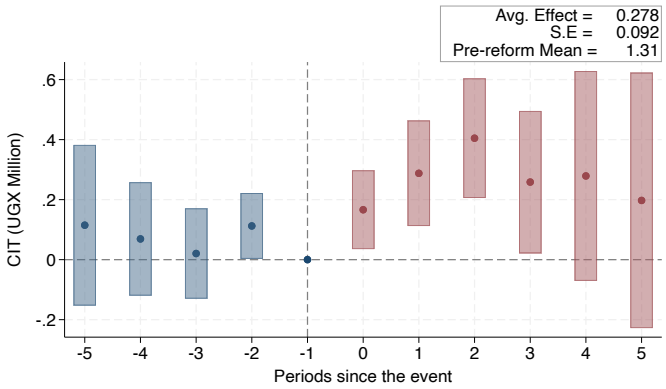


Figure A.3: Effects on Employee Compensation and Formal Employment: *Excluding 2009/10 Splits*

Notes: The figure presents the effects of district splitting on firms' employee compensation and formal employment, excluding 2009 and 2010 splits. The outcomes in Panels A and B are measured in local currency and expressed in constant 2016/17 prices, while Panel C reports the number of formal workers. Regressions are estimated on a balanced panel of firms and weighted by the logarithm of the pre-reform value of the corresponding outcome. The sample comprises 66,015 firm-year observations in Panel A and 42,678 firm-year observations in Panels B and C. The horizontal axis measures years relative to district splitting. Each dot denotes the point estimate of the treatment effect for a given year, with year −1 as the reference category, and the vertical bands denote the corresponding 95 percent confidence intervals. Coefficients are estimated using the staggered difference-in-differences estimator of Callaway and Sant'Anna (2021), with not-yet-treated units as the comparison group. The specification includes sector-year and accounting-month fixed effects. Standard errors are clustered at the firm level.

Panel A: Effects on CIT remitted



Panel B: Effective Tax Rate

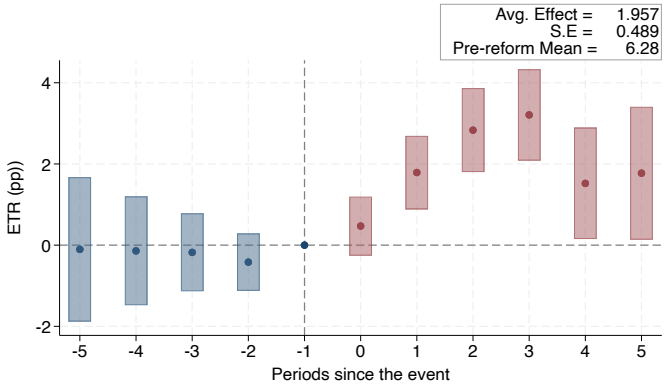
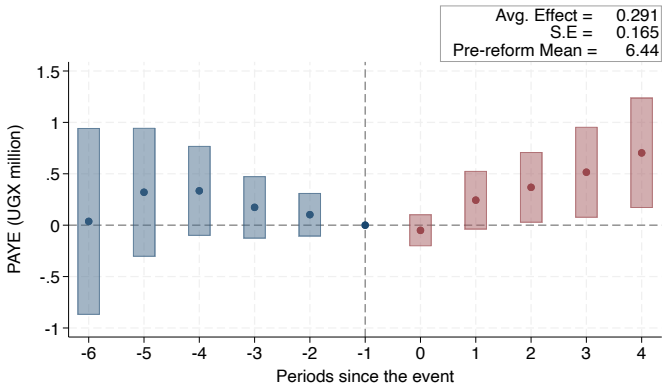


Figure A.4: Effects on Corporate Income Tax Payment: *Excluding 2009 and 2010 Splits*

Notes: The figure presents the effects of decentralization on corporate income tax remitted by firms (Panel A) and on the effective tax rate (Panel B), excluding 2009 and 2010 splits from the sample. The effective tax rate is computed as the ratio of tax paid to pre-tax profits and is set to missing for firms with negative profits, which are thereby dropped from the sample. Regressions are estimated on a balanced panel, with observations weighted by the logarithm of pre-reform revenues. The sample comprises 72,387 firm-year observations in Panel A and 61,730 in Panel B. Each dot denotes the point estimate of the treatment effect for a given year, with year −1 as the reference category, and the vertical bands denote the corresponding 95 percent confidence intervals. All specifications include sector-year and accounting-month fixed effects. Standard errors are clustered at the firm level.

Panel A: PAYE remitted



Panel B: CIT + PAYE remitted

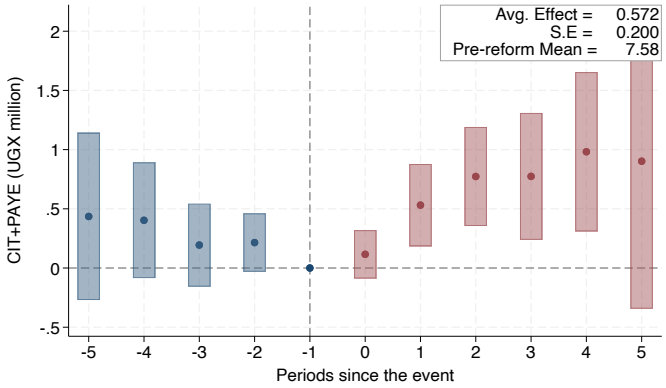


Figure A.5: Effects on Tax Remitted: *Excluding 2009/10 Splits*

Notes: The figure presents the effects of decentralization on PAYE tax (Panel A) and on the total tax remitted (Panel B), excluding firms in districts split in 2009 and 2010. Regressions are estimated on a balanced panel, with observations weighted by the logarithm of pre-reform revenues. The sample comprises 66,015 firm-year observations. Each dot denotes the point estimate of the treatment effect for a given year, with year −1 as the reference category, and the vertical bands denote the corresponding 95 percent confidence intervals. All specifications include sector-year and accounting-month fixed effects. Standard errors are clustered at the firm level.

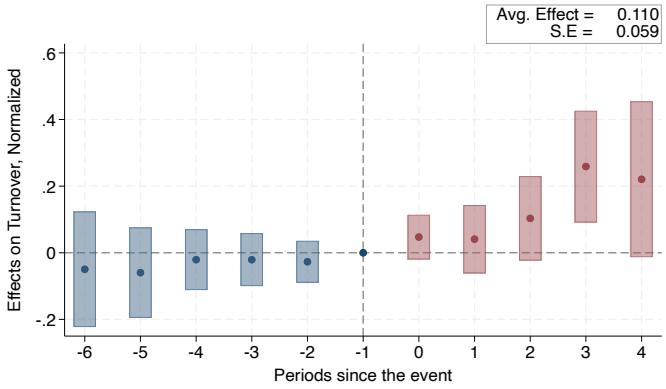
I.B. Excluding Kampala from the Comparison Group

Kampala constitutes an important part of the baseline comparison group but differs substantially from the rest of the country in both its economic structure and institutional environment. As Uganda's principal commercial center, Kampala has a greater concentration of large and formal firms, denser infrastructure, and substantially higher levels of economic activity. Consistent with these differences, the descriptive statistics in Table 1 show that firms in the baseline comparison group are, on average, larger than firms located in districts that are subsequently split, with these differences narrowing once Kampala is excluded.

These differences in levels do not, by themselves, invalidate the difference-in-differences design, which relies on parallel counterfactual trends rather than equality of pre-reform outcomes. A potential concern, however, is that firms in Kampala may experience systematically different changes over time because of economic or policy forces specific to the capital. If so, their inclusion in the comparison group could affect the estimated counterfactual evolution of firms in split districts. Moreover, given Kampala's economic importance and concentration of firms, the baseline estimates could disproportionately reflect comparisons with firms operating in an environment that is structurally different from that of the treated districts. The direction of any resulting bias is ambiguous and depends on how outcomes in Kampala would have evolved relative to those in the rest of the country.

To assess the sensitivity of the results to this concern, we re-estimate the main specifications after excluding all firms located in the Kampala Capital City Authority from the analysis sample. The resulting comparison group therefore consists only of firms located outside Kampala in districts that were not split during the study period. Appendix Figures A.6 through A.10 present the results for each of the main outcomes examined in the paper.The results are qualitatively similar to the baseline results in both their magnitudes and dynamic patterns. The persistence of the main findings in this more geographically comparable sample indicates that the estimated effects are not driven by the inclusion of Kampala in the comparison group.

Panel A: Turnover



Panel B: Gross Profits/Pre-reform Revenues

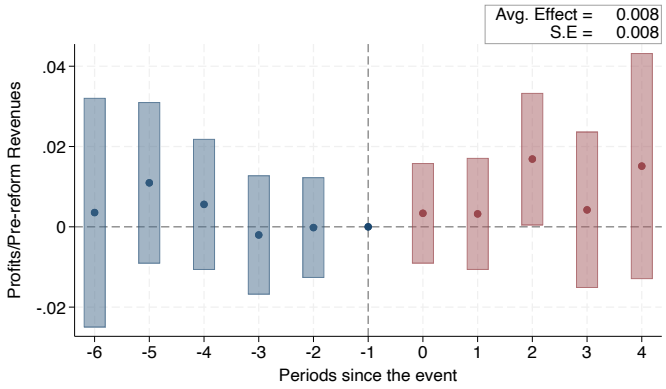
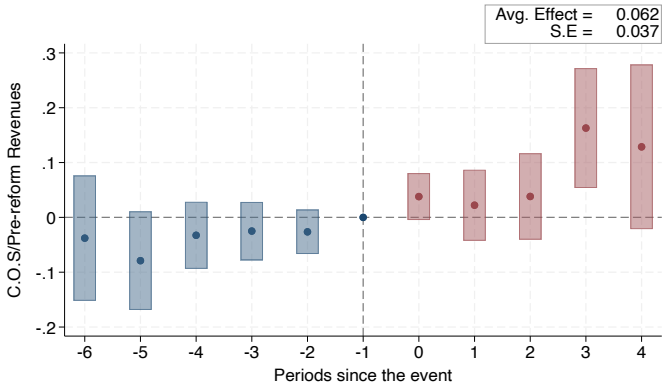


Figure A.6: Effects of Firms' Revenues and Profits: Excluding the Capital City

Notes: The figure shows the dynamic effects of district splitting on firms' turnover and profits, with the sample restricted to only firms outside the capital city. Turnover is normalized by its pre-reform value, while profits are normalized by pre-reform revenues. Coefficients in Panel A should therefore be interpreted as percentage changes, while those in Panels B should be interpreted as percentage point changes relative to pre-reform revenues. The horizontal axis measures years relative to district splitting. Each point represents the estimated treatment effect in that year, with year -1 as the reference category. Vertical bands indicate 90 percent confidence intervals. The specification includes fixed effects for sector of operation, pre-reform firm size, and accounting month. Standard errors are clustered at the firm level.

Panel A: Cost of Sales



Panel B: Raw Materials

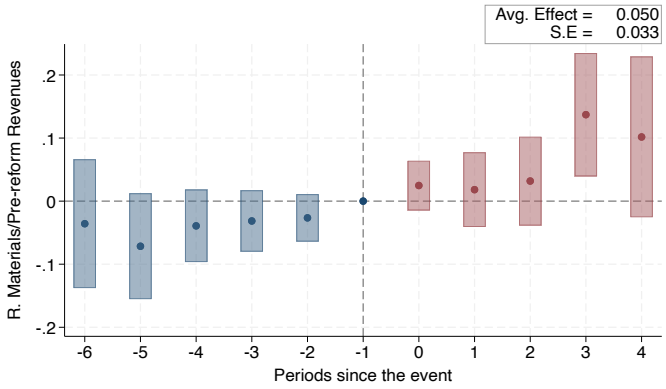


Figure A.7: Cost of Sales and Raw Materials: *Excluding Kampala*

Notes: The figure displays the dynamic effects of district splitting on firms' cost of sales and raw materials, excluding firms located in Kampala. Each outcome is scaled by its pre-reform values, implying the coefficients can be interpreted in percent changes. The horizontal axis measures years relative to district splitting. Each point represents the estimated treatment effect in that year, with year −1 as the reference category. Vertical bands indicate 95 percent confidence intervals. Coefficients are estimated using a staggered difference-in-differences approach following Callaway and Sant'Anna (2021), with not-yet-treated units as the comparison group. The specification includes fixed effects for sector of operation, pre-reform firm size, and accounting month. Standard errors are clustered at the firm level.

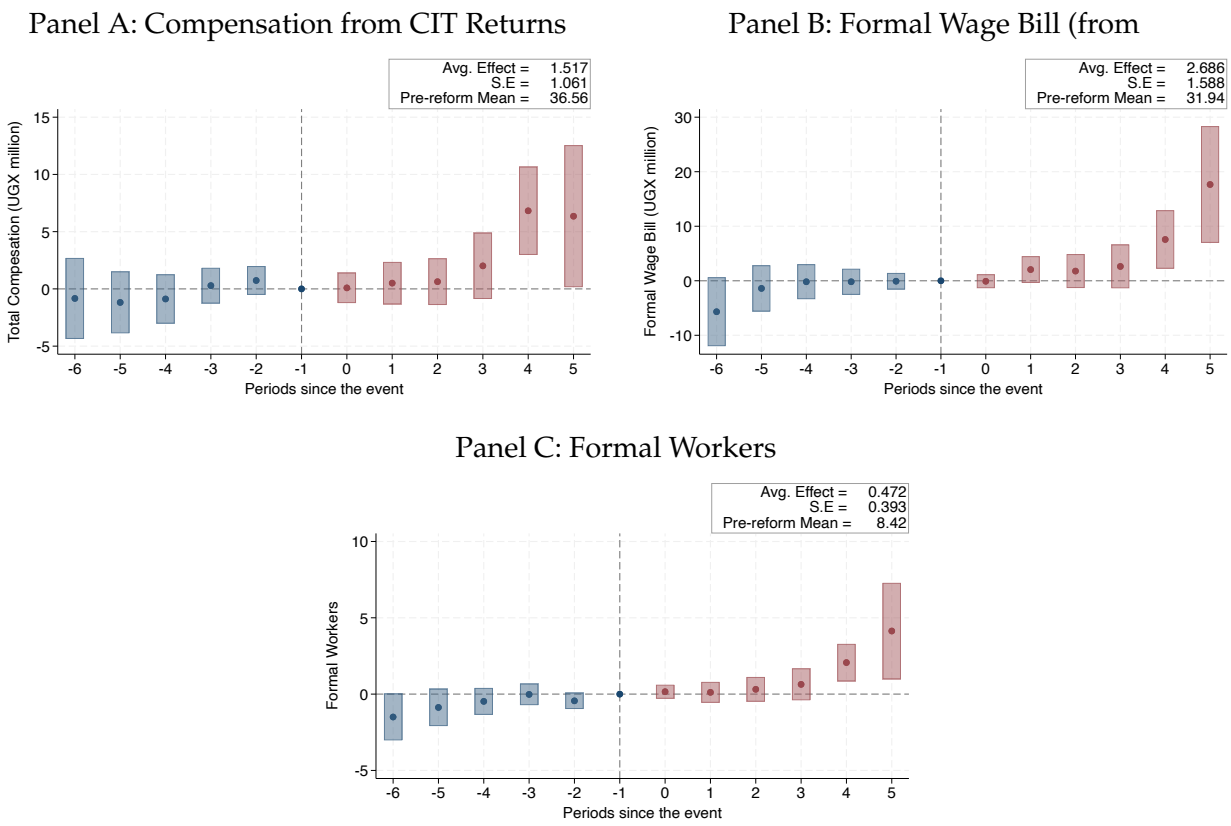
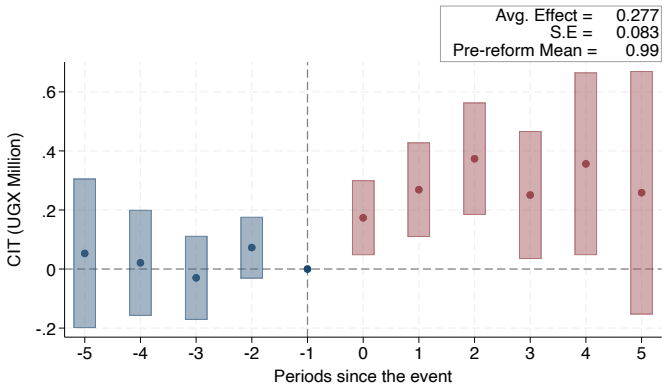


Figure A.8: Effects on Employee Compensation and Formal Employment: *Excluding Kampala*

Notes: The figure presents the effects of district splitting on firms' employee compensation and formal employment, excluding Kampala from the sample. The outcomes in Panels A and B are measured in local currency and expressed in constant 2016/17 prices, while Panel C reports the number of formal workers. Regressions are estimated on a balanced panel of firms and weighted by the logarithm of the pre-reform value of the corresponding outcome. The sample comprises 25,569 firm-year observations in Panel A and 14,301 firm-year observations in Panels B and C. The horizontal axis measures years relative to district splitting. Each dot denotes the point estimate of the treatment effect for a given year, with year −1 as the reference category, and the vertical bands denote the corresponding 95 percent confidence intervals. Coefficients are estimated using the staggered difference-in-differences estimator of Callaway and Sant'Anna (2021), with not-yet-treated units as the comparison group. The specification includes sector-year and accounting-month fixed effects. Standard errors are clustered at the firm level.

Panel A: Effects on CIT remitted



Panel B: Effective Tax Rate

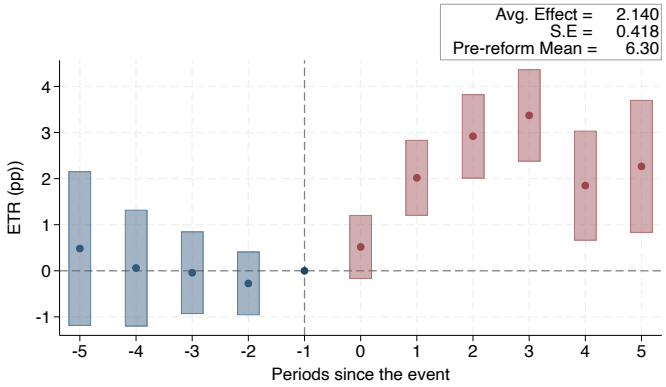
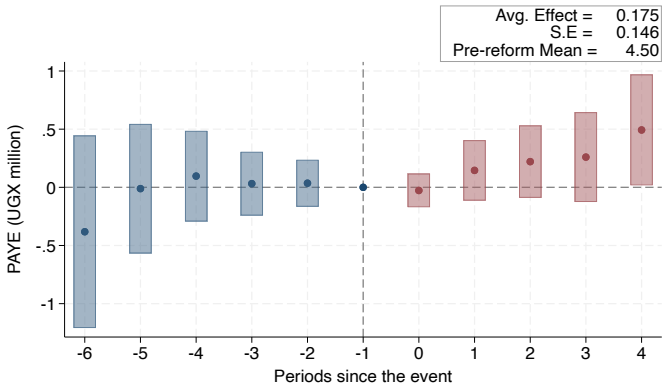


Figure A.9: Effects on Corporate Income Tax Payment: *Excluding Kampala*

Notes: The figure presents the effects of decentralization on corporate income tax remitted by firms (Panel A) and on the effective tax rate (Panel B), excluding Kampala from the sample. The effective tax rate is computed as the ratio of tax paid to pre-tax profits and is set to missing for firms with negative profits, which are thereby dropped from the sample. Regressions are estimated on a balanced panel, with observations weighted by the logarithm of pre-reform revenues. The sample comprises 27,864 firm-year observations in Panel A and 24,101 in Panel B. Each dot denotes the point estimate of the treatment effect for a given year, with year −1 as the reference category, and the vertical bands denote the corresponding 95 percent confidence intervals. All specifications include sector-year and accounting-month fixed effects. Standard errors are clustered at the firm level.

Panel A: PAYE remitted



Panel B: CIT + PAYE remitted

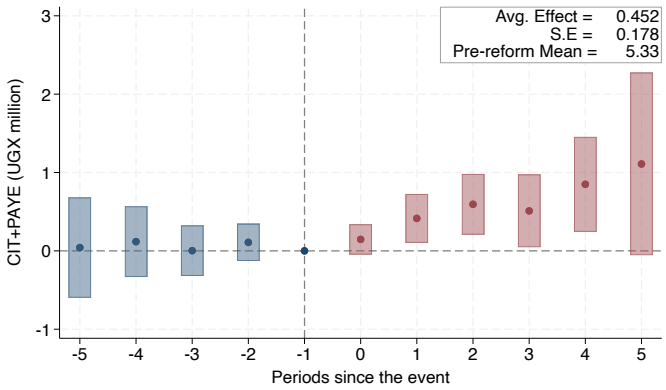
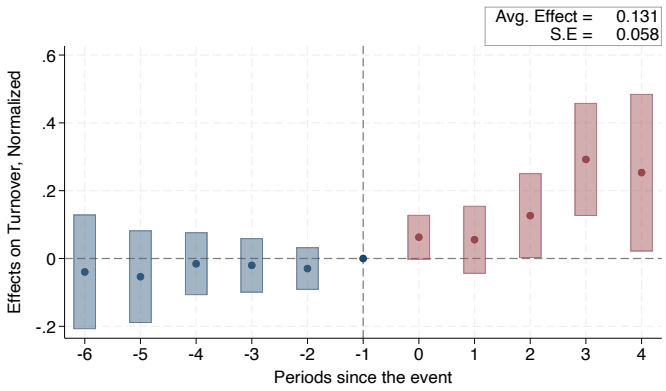


Figure A.10: Effects on Tax Remitted: *Excluding Kampala*

Notes: The figure presents the effects of decentralization on PAYE tax (Panel A) and on the total tax remitted (Panel B), excluding firms located in Kampala. Regressions are estimated on a balanced panel, with observations weighted by the logarithm of pre-reform revenues. The sample comprises 25,569 firm-year observations. Each dot denotes the point estimate of the treatment effect for a given year, with year −1 as the reference category, and the vertical bands denote the corresponding 95 percent confidence intervals. All specifications include sector-year and accounting-month fixed effects. Standard errors are clustered at the firm level.

II. ADDITIONAL FIGURES

Panel A: Turnover



Panel B: Gross Profits/Pre-reform Revenues

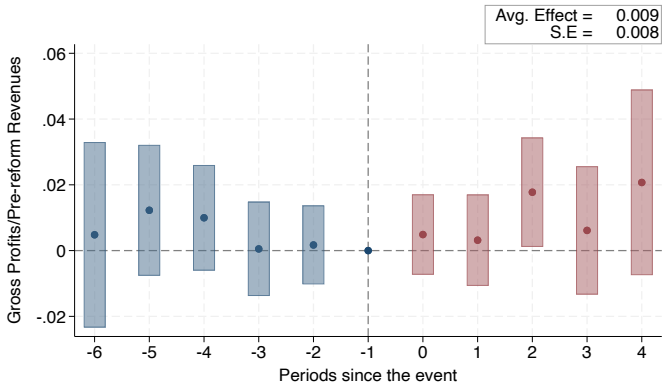
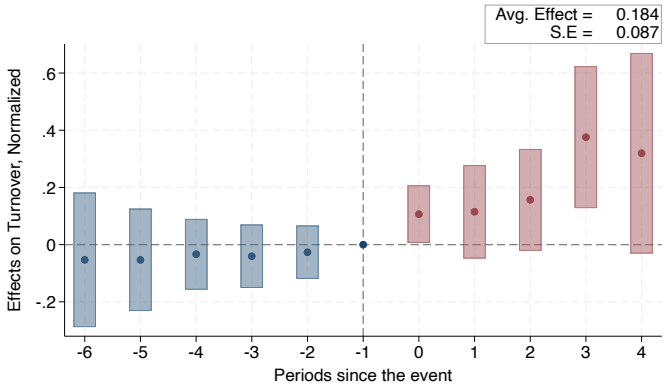


Figure A.11: Effects of Firms' Revenues and Profits: Never-Treated as the Comparison Group

Notes: The figure shows the dynamic effects of district splitting on firms' turnover and profits, with never treated units used as the comparison group. Turnover is normalized by its pre-reform value, while profits are normalized by pre-reform revenues. Coefficients in Panel A should therefore be interpreted as percentage changes, while those in Panels B should be interpreted as percentage point changes relative to pre-reform revenues. The horizontal axis measures years relative to district splitting. Each point represents the estimated treatment effect in that year, with year -1 as the reference category. Vertical bands indicate 95 percent confidence intervals. The specification includes fixed effects for sector of operation, pre-reform firm size, and accounting month. Standard errors are clustered at the firm level.

Panel A: Turnover



Panel B: Gross Profits/Pre-reform Revenues

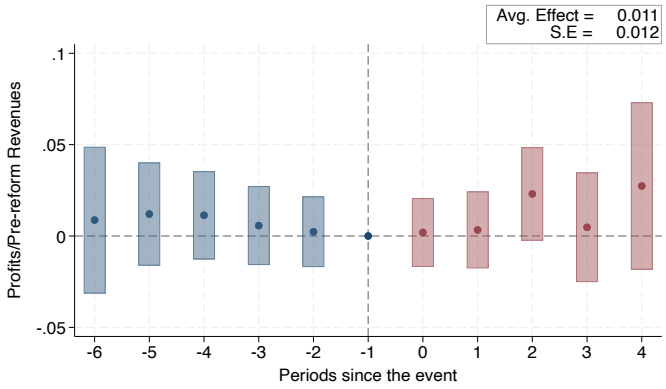


Figure A.12: Effects of Firms' Revenues and Profits: Winsorizing at 97.5th Percentile

Notes: The figure shows the dynamic effects of district splitting on firms' turnover and profits. Turnover is normalized by its pre-reform value, while profits are normalized by pre-reform revenues. All outcomes are winsorized at the 97.5th percentile. Coefficients in Panel A should be interpreted as percentage changes, while those in Panels B should be interpreted as percentage point changes relative to pre-reform revenues. The horizontal axis measures years relative to district splitting. Each point represents the estimated treatment effect in that year, with year -1 as the reference category. Vertical bands indicate 90 percent confidence intervals. Coefficients are estimated using a staggered difference-in-differences approach following Callaway and Sant'Anna (2021), with not-yet treated units as the comparison group. The specification includes fixed effects for sector of operation, pre-reform firm size, and accounting month. Standard errors are clustered at the firm level.

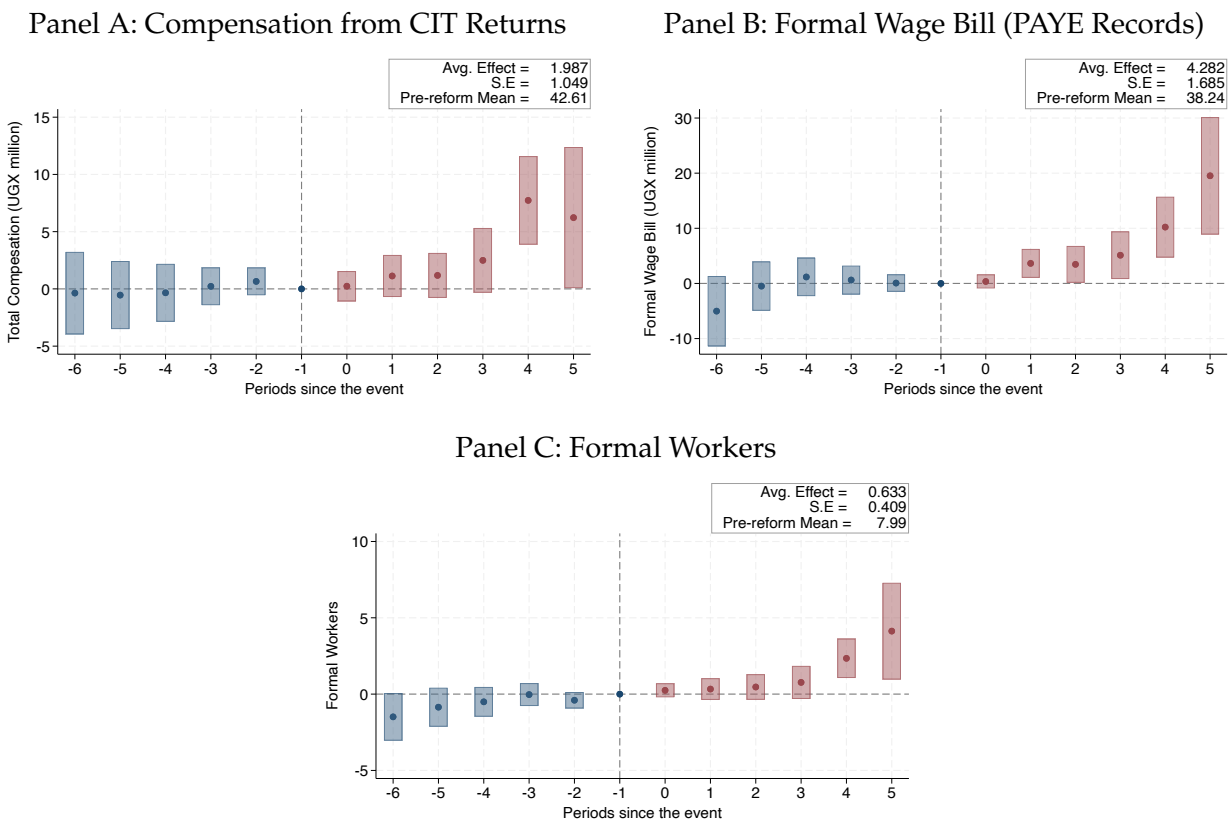


Figure A.13: Effects on Employee Compensation and Formal Employment: *Never-Treated as Comparison*

Notes: The figure presents the effects of district splitting on firms' employee compensation and formal employment. The outcomes in Panels A and B are measured in local currency and expressed in constant 2016/17 prices, while Panel C reports the number of formal workers. Regressions are estimated on a balanced panel of firms and weighted by the logarithm of the pre-reform value of the corresponding outcome. The sample comprises 73,098 firm-year observations in Panel A and 46,305 firm-year observations in Panels B and C. The horizontal axis measures years relative to district splitting. Each dot denotes the point estimate of the treatment effect for a given year, with year −1 as the reference category, and the vertical bands denote the corresponding 95 percent confidence intervals. Coefficients are estimated using the staggered difference-in-differences estimator of Callaway and Sant'Anna (2021), with not-yet-treated units as the comparison group. The specification includes sector-year and accounting-month fixed effects. Standard errors are clustered at the firm level.

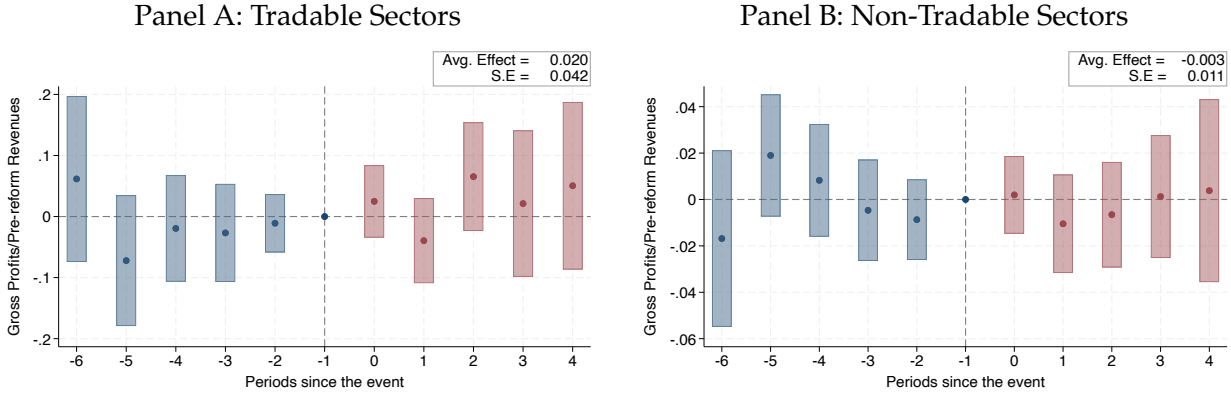
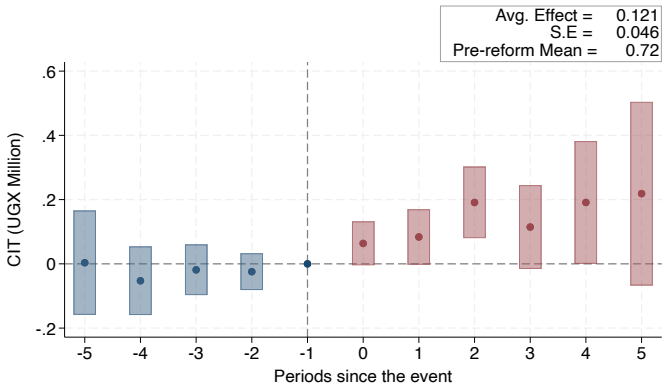


Figure A.14: Effects on Profits: Tradable vs Non-tradable

Notes: The figure presents the effects of the reform on firm profitability separately for firms in tradable and non-tradable sectors. Tradable sectors include agriculture, mining and quarrying, manufacturing, and electricity and gas production. Non-tradable sectors include water supply and sewerage management, construction, retail trade, transport and storage, accommodation and food services, real estate, professional services, and administrative and support services. The classification excludes sectors that cannot be cleanly categorized as either tradable or non-tradable, such as wholesale trade. The sample comprises 7,083 firm-year observations in Panel A and 25,884 firm-year observations in Panels B. The horizontal axis measures years relative to district splitting. Each dot denotes the point estimate of the treatment effect for a given year, with year −1 as the reference category, and the vertical bands denote the corresponding 95 percent confidence intervals. Coefficients are estimated using the staggered difference-in-differences estimator of Callaway and Sant'Anna (2021), with not-yet-treated units as the comparison group. The estimations condition on the sector of operation and the pre-reform accounting. Standard errors are clustered at the firm level.

Panel A: Effects on CIT remitted



Panel B: Effective Tax Rate

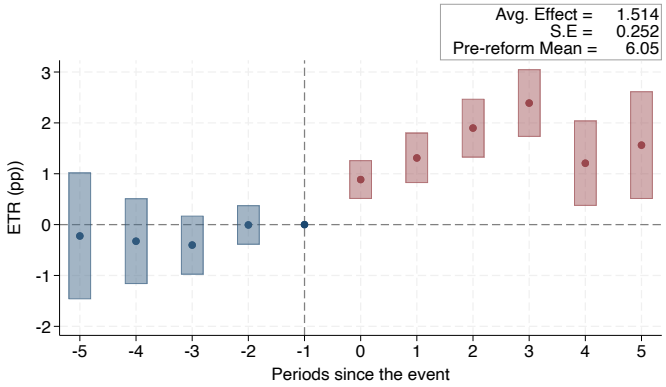


Figure A.15: Effects on Corporate Income Tax Payment: *Unbalanced Panel*

Notes: The figure presents the effects of decentralization on corporate income tax remitted by firms (Panel A) and on the effective tax rate (Panel B). The effective tax rate is computed as the ratio of tax paid to pre-tax profits and is set to missing for firms with negative profits, which are thereby dropped from the sample. Regressions are estimated on an unbalanced panel, with observations weighted by the logarithm of pre-reform revenues. The sample comprises 199,911 firm-year observations in Panel A and 153,904 in Panel B. Each dot denotes the point estimate of the treatment effect for a given year, with year −1 as the reference category, and the vertical bands denote the corresponding 95 percent confidence intervals. All specifications include sector-year and accounting-month fixed effects. Standard errors are clustered at the firm level.

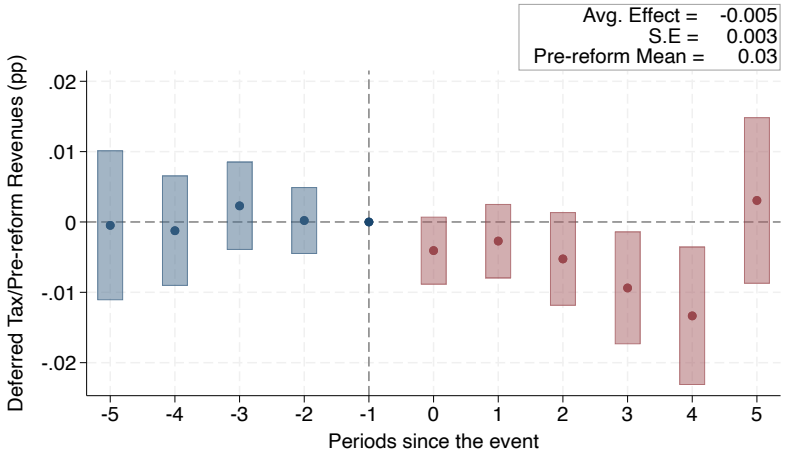


Figure A.16: Effects on Deferred Tax Liability

Notes: The figure presents the effect of decentralization on deferred tax liabilities, scaled by pre-reform revenues and expressed in percentage points. The regression is estimated on a balanced panel of firms and weighted by the logarithm of pre-reform revenues. The sample comprises 68,742 firm-year observations. Each dot denotes the point estimate of the treatment effect for a given year, with year −1 as the reference category, and the vertical bands denote the corresponding 95 percent confidence intervals. The specification includes sector-year and accounting-month fixed effects. Standard errors are clustered at the firm level.

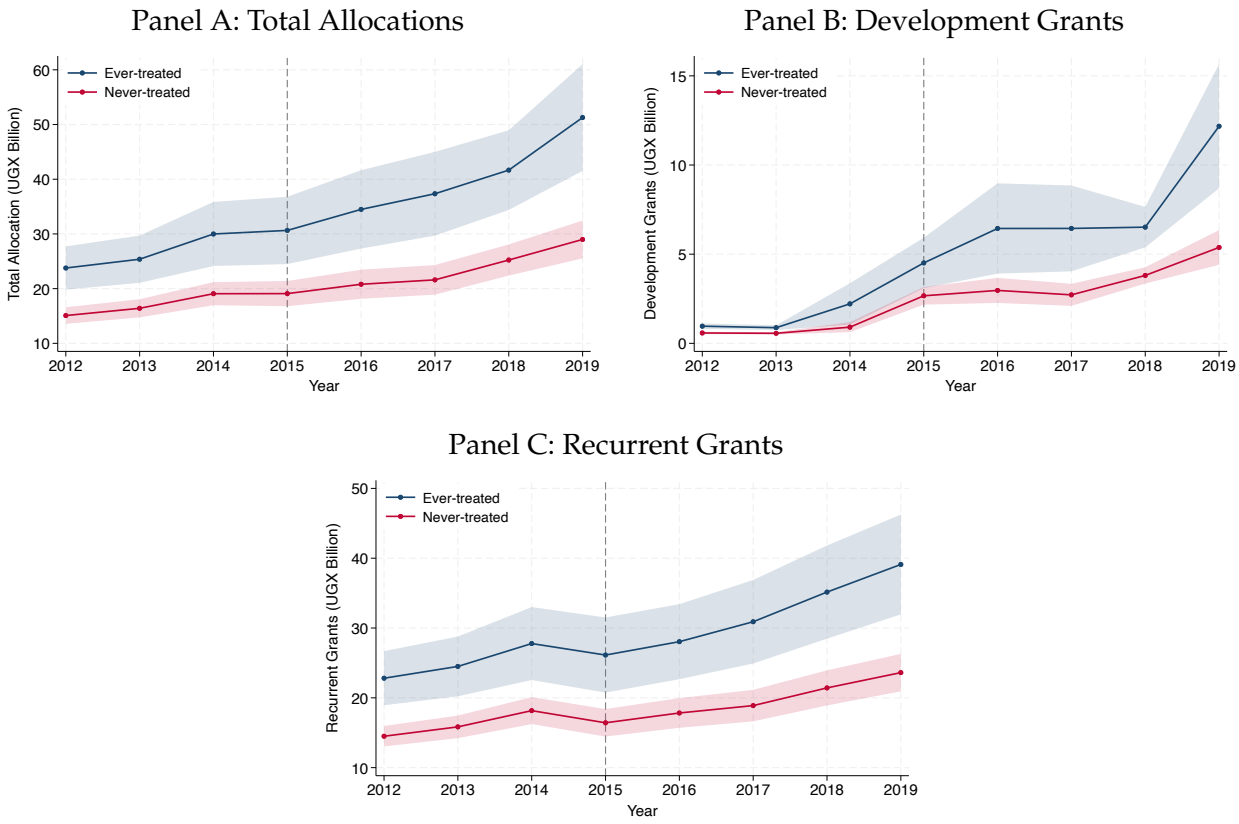


Figure A.17: Intergovernmental Allocations: Raw Data

Notes: The figure presents raw trends in central government intergovernmental allocations to districts by treatment status. Panel A shows total allocations, Panel B shows development grants, and Panel C shows recurrent grants, all in billions of Uganda shillings. Ever-treated districts are those subsequently split during or after the 2015/16 wave of district creation; never-treated districts are those not split during the sample period. For each year, we compute the mean allocation across districts within each group, aggregating allocations at the level of the original parent district so that pre- and post-split values reflect the same set of territories. Solid lines denote group means and shaded bands denote 95 percent confidence intervals. The years on the horizontal axis indicate the starting period of the fiscal year (e.g., 2015 represents fiscal year 2015/2016) Data on district-level allocations are drawn from the Ministry of Finance, Planning and Economic Development.

III. ADDITIONAL TABLES